
CISCO CERTIFIED NETWORK ASSOCIATE

CCNA 200-301 VERSION 2.0

Notes, Configurations and Labs

WRITTEN FROM ZERO

No prior networking assumed. Protocol theory and device configuration taught together, because the exam tests both and mostly tests the second.

 25% Infrastructure and Connectivity	 25% Switching and Network Access	 20% IP Routing	 20% Services and Security	 10% AI, Operations and Management
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Fifty-five per cent of this exam is hands-on. Of the twenty-nine topics Cisco publishes, eight begin *Troubleshoot* or *Diagnose* and eight begin *Configure*. Only six begin *Describe*. Under the previous blueprint, v1.1, the word “troubleshoot” appeared in no topic statement at all. You cannot pass this exam by reading.

Aligned to blueprint version 2.0, live from 3 February 2027 • August 27, 2026

The Exam, in One Page

Exam	Implementing and Administering Cisco Solutions, 200-301 CCNA, blueprint version 2.0
Length	120 minutes
Live from	3 February 2027. Version 1.1 retires on 2 February 2027; the exam number does not change, so check which blueprint a book or course is written against before you trust it
Domains	Five: Network Infrastructure and Connectivity (25%), Switching and Network Access (25%), IP Routing (20%), Network Services and Security (20%), AI and Network Operations and Management (10%)
Topics	Twenty-nine, numbered 1.1 to 5.6. Every chapter of this volume is tagged with the ones it discharges, and Appendix D proves all twenty-nine are covered
The shape of it	Eight topics ask you to <i>troubleshoot</i> or <i>diagnose</i> ; eight ask you to <i>configure</i> ; six ask you to <i>describe</i> ; the rest ask you to interpret, validate, manage, select or use. Roughly a third of the exam is fixing something broken at a command line, under time pressure
Agentic AI	Cisco’s own exam description warns that you “may be required to evaluate output and recommendations from agentic AI and digital network assistants” — and not only in domain 5. See Chapter 32

What changed from version 1.1, and why it matters to you

If you have an older book, it is teaching a different exam. QoS, NTP, TFTP and FTP, the OSI and TCP/IP model descriptions, TCP versus UDP comparison, device-role and topology-architecture theory, WLC GUI configuration, REST APIs, JSON and Terraform have all gone. OSPFv3, hypervisors and containers, SFTP and SCP, DNS record diagnosis, storm control, RA guard, PoE configuration, DHCP server configuration and switch virtual interfaces have arrived — along with a whole domain on AI in network operations. Above all, troubleshooting went from nothing to about a third of the paper.

How to Use This Volume

This book assumes you have never configured a network device and have not studied networking before. It teaches the theory and the configuration together, in the order they can actually be learned, and every chapter states which exam topics it discharges.

The furniture in every chapter

Each chapter is built from the same parts, always in the same order, so you always know where you are.

Element	What it is for
EXAM TOPICS 2.1	The blueprint topics this chapter discharges, colour-coded by domain. If you are revising against the blueprint, navigate by these
🎯 Learning Outcomes	What you will be able to <i>do</i> after the chapter, written as actions
🔗 Before You Start	The specific earlier chapters this one leans on
🔑 Key Terms	Vocabulary introduced here. All terms also appear in Appendix F
📖 Definition	A precise statement of a term
💡 Concept	The intuition behind it — why it works this way
>_ Configure	The actual commands, in the order you type them. Read these with a device or simulator open, not in an armchair
🔍 Verification	The show command that proves the configuration worked, and the output you should see. Learning to read this output is learning to troubleshoot
❗ Common Pitfalls	Mistakes made every cohort
🔧 Break & Fix	A broken device to diagnose. This is the 28% of the exam that is troubleshooting, practised in every chapter rather than revised once at the end
🧪 Lab	A hands-on exercise with a topology, an addressing table, numbered tasks and a verification step. Labs are numbered and indexed in Appendix G
🎓 On the exam	How this material is actually tested, and the traps in the question wording
❓ Checkpoint	Quick self-tests. Answers in Appendix E
📋 Chapter Summary	The chapter compressed to what must be retained
✍ Review Questions	Exam-style questions for practice

How to study with it

1. **Type every configuration.** Not copy and paste — type it. The exam asks you to produce commands from memory under time pressure, and muscle memory is the only thing that survives the pressure.
2. **Never skip the verification block.** Half of troubleshooting is knowing what healthy output looks like, and you only learn that by looking at it repeatedly when nothing is wrong.

3. **Do the Break & Fix before reading the answer.** Sit with the symptom. The exam will not hand you a diagnosis.
4. **Build the labs, do not read them.** A lab you have read is worth nothing on an exam that asks you to configure.
5. **Use Appendix D to revise.** It maps every one of the twenty-nine exam topics to the chapter that covers it, so you can check your own coverage rather than hoping.

What you need

- **Parts I to III: Cisco Packet Tracer.** Free with a Cisco Networking Academy account, runs on a modest laptop, and covers everything in the first three parts — VLANs, trunking, EtherChannel, spanning tree, static routing and OSPF — perfectly well.
- **Parts IV onwards: CML or GNS3.** From Part IV the exam names real platform behaviour — topic [4.3](#) says *IOS XE* explicitly, and SFTP, SCP and Ansible need a real control plane. Packet Tracer approximates these; it does not reproduce them.
- **A notebook.** Subnetting is done on paper in the exam room.

About the configurations in this book

Every configuration here is written to be syntactically correct for Cisco IOS and IOS XE, and every one follows the command order a device actually requires. They have **not** been verified on live hardware. Platform, image and version differences are real: a command that works on one switch may be unavailable or spelled differently on another. Before issuing a lab to students, run it once yourself on whatever platform your department has. That advice is worth more than a claim of validation this book cannot honestly make.

Syllabus

Course description

A first course in computer networking, taught to certification standard. It develops the ability to design, configure, verify and troubleshoot a small enterprise network — switching, routing, addressing, services and security — and prepares the student for the Cisco CCNA 200-301 v2.0 examination. No prior networking study is assumed. The course is built around laboratory work, because the certification it targets is more than half practical.

Course learning outcomes

On completion, a student will be able to:

CLO	Outcome	Exam domain
CLO-1	Explain how data moves across a switched and routed network, and use addressing, subnetting and the transport layer correctly. (C2, C3)	1.3 1.4
CLO-2	Configure switching for a small enterprise: VLANs, trunks, EtherChannel, spanning tree and edge-host connectivity. (C3)	2.1 2.2 2.5
CLO-3	Configure and interpret IP routing — static routes, single-area OSPF for IPv4 and IPv6, and first hop redundancy. (C3, C4)	3.1 3.2 3.3 3.4
CLO-4	Configure network services and security: DHCP, NAT, access control lists, device access control and layer 2 protection. (C3)	4.1 4.3 4.6 4.7
CLO-5	Diagnose and repair a broken network from symptoms, using <code>show</code> output, ping, traceroute and packet capture. (C4, C5)	2.4 1.6 1.7
CLO-6	Describe modern network operations — management approaches, SNMP, syslog, automation with Ansible, and the role of agentic AI — and select an effective prompt for an operations task. (C2, C3)	5.1 5.2 5.3 5.5
CLO-7	Work safely and professionally on shared infrastructure, documenting changes and respecting the blast radius of a configuration. (A3)	—

CLO-5 carries the heaviest assessment weight deliberately: it is the outcome the exam weights most heavily and the one students are least prepared for.

Assessment

Component	Weight	Notes
Laboratory portfolio	30%	One submission per lab: topology, running configuration, and the <code>show</code> output proving it works

Component	Weight	Notes
Break & Fix assessments	25%	Timed. A pre-broken topology; find and fix the fault, and write down what it was. Mirrors the exam's heaviest verb
Configuration midterm	15%	At a device or simulator, not on paper
Theory examination	20%	Written, covering the <i>describe</i> and <i>interpret</i> topics
Professional practice	10%	Documentation quality, change discipline, teamwork in the lab

A department running this course inside a degree can map the laboratory portfolio and the Break & Fix assessments onto a practical component, and the remaining 35% onto the written component, without changing the weights.

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PART I

Foundations

Before you can configure a network you have to know what is moving through it, how it is addressed, and how to talk to a device at all. Part I builds that foundation from nothing.

It is also where two of the exam's heaviest habits are formed. The first is **subnetting on paper**, quickly and without a calculator — topic [1.3](#) asks you to troubleshoot addressing, and you cannot troubleshoot what you cannot compute. The second is **reading device output**, which is what topic [2.4](#) tests and what every troubleshooting question in the paper depends on.

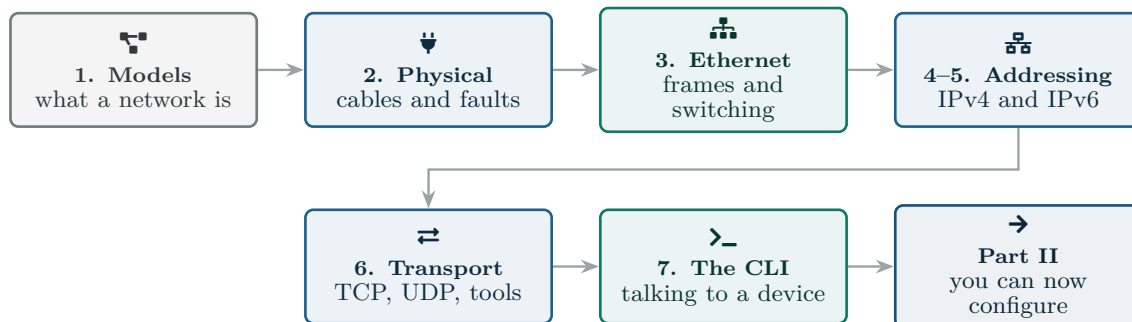


Figure P1.1: Part I in order. Each chapter is a prerequisite for the next, and the last two are the ones everything in Parts II to V depends on.

Networks, Models and This Exam

🎯 Learning Outcomes

By the end of this chapter you will be able to:

- **Describe** what a network does, in terms of the four things every network must solve: naming, addressing, forwarding and error handling.
- **Explain** the layered model well enough to use it as a troubleshooting tool, which is the only reason this exam still cares about it.
- **Identify** the devices in a small enterprise network and say what each one decides.
- **Navigate** the CCNA v2.0 blueprint and this book against it.

🔗 Before You Start

Nothing. This is the first chapter, and the course assumes no prior networking.

📖 Key Terms in This Chapter

host • node • link • frame • packet • segment • encapsulation • layer • protocol • switch • router • access layer • distribution layer • broadcast domain • collision domain

1.1 What a network has to solve

Two computers in the same room, joined by a cable, can exchange bytes. Everything else in networking exists because that arrangement does not scale. Add a third machine and you need a way to say *which* machine you mean. Add a building and you need a way to find a machine you cannot see. Add the internet and you need a way to reach a machine whose owner you have never met, over equipment owned by people you will never meet either.

💡 The four problems, and the four answers

Every network technology in this book is an answer to one of four questions.

- **Who am I talking to?** → addressing. MAC addresses locally (Chapter 3), IP addresses globally (Chapter 4).
- **How do I find them?** → forwarding. Switches learn (Chapter 3); routers decide (Chapter 15).
- **How do we not talk over each other?** → media access and loop prevention (Chapter 3, Chapter 12).
- **What if it goes wrong?** → error detection, acknowledgement and retransmission (Chapter 6).

When you meet a new protocol, ask which of the four it is answering. It will be one of them.

1.2 Layers, and why they survive

A network stack is split into layers so that each layer can be changed without rewriting the others. Wi-Fi replaced the cable without anyone rewriting the web. That is the engineering argument, and it is real, but it is not why the layered model matters to you this year.

💡 The model is a troubleshooting tool

When something does not work, the layered model tells you *what to test next*. If a cable is unplugged, checking DNS is a waste of a minute you do not have in the exam. Work up: is the link up? Does it have an address? Can it reach its gateway? Can it resolve a name? Every troubleshooting question in this exam — and there are eight topics' worth — is answered by walking the layers in order.

The v2.0 blueprint no longer asks you to *describe* the OSI model, as v1.1 did. It assumes you use it.

#	Layer	What it decides	What breaks, and the symptom
7–5	Application	What the data means	Wrong URL, expired certificate; “the site is down but ping works”
4	Transport	Which program on the host, and whether loss is repaired	Blocked port; “it pings but the application times out”
3	Network	Which network, and the path between networks	Wrong IP, wrong mask, no route; “I can reach my own subnet but nothing else”
2	Data link	Which device on this link	Wrong VLAN, trunk down, spanning tree blocking; “the port is up but nothing passes”
1	Physical	Bits on the wire	Cable, transceiver, duplex or speed mismatch; “the port is down”, or worse, up with errors climbing

The bottom two rows are where this exam starts: topic [1.1](#) is entirely physical-layer diagnosis, and it is examined in Chapter 2.

📄 Encapsulation, and the names of things

As data moves down the stack each layer wraps it in a header. The wrapped unit gets a different name at each layer, and using the right one is how you show you know which layer you are discussing.

- Application data becomes a **segment** at layer 4 (TCP) or a **datagram** (UDP), when a port number is attached.
- That becomes a **packet** at layer 3, when IP addresses are attached.
- That becomes a **frame** at layer 2, when MAC addresses are attached.
- That becomes **bits** at layer 1.

Switches forward frames. Routers forward packets. That single sentence explains most of the difference between the two devices.

1.3 The devices, and what each one decides

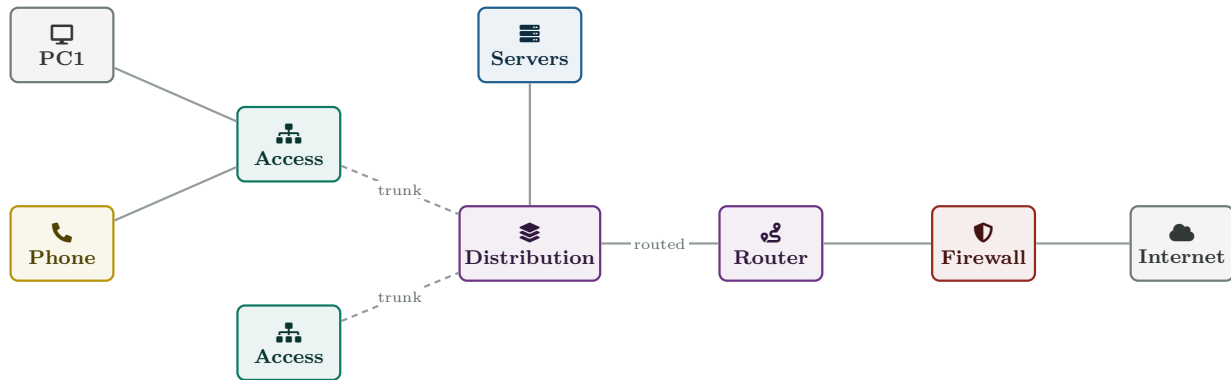


Figure P1.1: A small enterprise network. Access switches connect end devices; a distribution switch aggregates them and routes between VLANs; the router reaches other networks; the firewall polices what leaves. Every chapter in Parts II to IV configures one piece of this picture.

Device	The decision it makes	Where in this book
Switch (layer 2)	“Which port leads to this MAC address?” Learns by watching source addresses; floods when it does not know	Chapter 3, then all of Part II
Router	“Which interface leads to this IP network?” Never floods; drops what it cannot route	Part III
Multilayer switch	Both. Switches within a VLAN, routes between them using an SVI	Chapter 10, Chapter 15
Access point	Bridges wireless clients onto a wired VLAN	Chapter 27
Firewall	“Is this flow permitted?” Stateful, unlike an access list	Chapter 24, Chapter 26

Two domains worth keeping straight

A **collision domain** is the set of ports that could transmit at the same instant and interfere. Every switch port is its own collision domain, which is why modern switched networks do not collide — and why a **collisions** counter that is climbing means something is badly wrong, usually a duplex mismatch (Chapter 2).

A **broadcast domain** is the set of devices that receive each other’s broadcasts. One VLAN is one broadcast domain. Splitting broadcast domains is the entire purpose of Chapter 8, and joining them again is the purpose of Chapter 15.

1.4 This exam, and this book

How the blueprint is built, and how to read it

Cisco publishes twenty-nine numbered topics in five domains. The number is the whole contract: if a technique is not in a topic statement, it is not directly examinable, and if it is, the *verb* tells you the depth expected.

- *Describe* — you must be able to explain it. Six topics.

- *Interpret, Validate, Select, Use, Manage* — you must be able to read output or make a choice. Seven topics.
- *Configure* — you must produce the commands. Eight topics.
- *Troubleshoot* or *Diagnose* — you must find and fix a fault. Eight topics.

Sixteen of twenty-nine are configure-or-fix. That is the single most useful fact about this exam, and it should change how you study: time at a command line beats time with a highlighter, by a wide margin.

Every chapter of this book carries its topic numbers under the title. If you want to revise by blueprint rather than by chapter, Appendix D maps all twenty-nine topics to the chapters that cover them.

! Common Pitfalls

- **Studying a v1.1 book.** The exam number, 200-301, did not change when the blueprint did. A book that teaches QoS, NTP, REST APIs, JSON or Terraform in a CCNA context is written for the old blueprint. Check the version before you trust the contents.
- **Learning the OSI layers as a list to recite.** Nobody will ask you to name layer 6. They will show you a broken network and expect you to work upwards.
- **Assuming “switch” means layer 2.** Multilayer switches route. Most of the distribution layer in a real network is switches doing routing, and topic 2.1 explicitly includes switch virtual interfaces.

? Checkpoint — Test Yourself

1. A user reports that a web page will not load. Ping to the server’s IP address succeeds. Which layers have you just ruled out, and which is the first one you should now suspect?
2. A switch port shows as up, but the collisions counter is increasing steadily. Which layer is at fault, and what is the most likely single cause?
3. Why does a router drop a packet for an unknown network when a switch floods a frame for an unknown MAC address?

✓ Chapter Summary

- Networks solve four problems: addressing, forwarding, media access and error handling. Every protocol you meet answers one of them.
- Layers matter because they tell you what to test next. Work upwards from the physical layer; do not start at DNS.
- Switches forward frames using MAC addresses and flood when they do not know. Routers forward packets using IP addresses and drop when they do not know.
- One VLAN is one broadcast domain. Every switch port is its own collision domain.
- CCNA v2.0 has twenty-nine topics. Sixteen of them ask you to configure or to fix. Study accordingly.

🔧 Review Questions

1. (MCQ) Which unit of data carries MAC addresses? (a) segment (b) packet (c) frame (d) datagram
2. (MCQ) A device receives a frame for a destination MAC address it has never seen. What does it do? (a) drops it (b) floods it out all ports in the VLAN except the one it arrived on (c) returns it to the sender (d) routes it

3. **(Short)** Give one reason the v2.0 blueprint dropped the requirement to describe the OSI model, and say what it expects instead.
4. **(Short)** Explain the difference between a collision domain and a broadcast domain, and give the device that bounds each.
5. **(Applied)** A colleague says “we do not need to learn troubleshooting, it is only one section of the exam”. Using the verb counts in the blueprint, explain why that is wrong.

The Physical Layer and Cable Faults

EXAM TOPICS **1.1**

🎯 Learning Outcomes

By the end of this chapter you will be able to:

- **Select** the right medium — copper, multimode fibre, single-mode fibre — for a given distance and speed.
- **Read** the interface counters that expose a physical fault, and say which counter implicates which cause.
- **Diagnose** the classic faults: duplex mismatch, speed mismatch, wrong cable type, bad pin-out, dirty or over-long fibre, and a transceiver operating outside its optical budget.
- **Distinguish** a port that is down from a port that is up and damaging traffic — the second is far more dangerous.

🔗 Before You Start

Chapter 1 for the layered model and the idea of a collision domain.

📖 Key Terms in This Chapter

twisted pair • Cat5e/Cat6 • multimode • single-mode • SFP • attenuation • auto-negotiation • duplex mismatch • CRC error • runt • giant • late collision • input error • pin-out • straight-through • crossover • Auto-MDIX

🎓 Read the topic wording

Topic **1.1** says *diagnose* interface and cable issues, and then names them: “collisions, errors, mismatched duplex, speed, distance, interface, signal levels, pin out, and cable types”. That list is a syllabus in itself. You will not be asked to describe a cable; you will be shown counters and asked what is wrong.

2.1 Choosing the medium

Medium	Typical reach	Speed	Use it when
Cat5e twisted pair	100 m	1 Gbps	Desk to access switch. The default, and the distance limit is a hard 100 m including patch leads
Cat6 / Cat6a	100 m (55 m for 10G on Cat6)	1–10 Gbps	Uplinks within a building, or any run you expect to upgrade
Multimode fibre (OM3/OM4)	300–400 m	1–10 Gbps	Between floors or between buildings on one campus. Cheaper optics, shorter reach
Single-mode fibre	Kilometres	1–100 Gbps	Between sites. More expensive optics, effectively unlimited campus reach

Medium	Typical reach	Speed	Use it when
Direct-attach copper	1–10 m	10 Gbps+	Switch to switch inside one rack

💡 Distance is a fault, not a preference

The exam names *distance* as a diagnosable issue for a reason. Exceed the reach of a medium and the link does not fail cleanly — it comes up and then corrupts frames, which shows as climbing CRC errors rather than a down port. A 95 m run that works in winter and fails in summer is a real phenomenon, and the answer is fibre, not another patch lead.

2.1.1 Copper pin-outs, and why you can mostly forget them

A **straight-through** cable connects pin 1 to pin 1; a **crossover** swaps the transmit and receive pairs. Historically you needed a crossover between two devices of the same kind — switch to switch, PC to router — and a straight-through between unlike devices.

Auto-MDIX makes the switch detect and correct this itself, and it is on by default on every modern platform. It is still examinable because it fails in two situations: when a port has been hard-coded to a speed and duplex (Auto-MDIX requires auto-negotiation to be on), and on older or simulated gear. If a link between two switches will not come up and both ends are hard-coded, the pin-out is a live suspect.

2.2 The counters that name the fault

❏ SW1# show interfaces GigabitEthernet0/1

```
GigabitEthernet0/1 is up, line protocol is up (connected)
Hardware is Gigabit Ethernet, address is 00d0.58a1.0201
MTU 1500 bytes, BW 1000000 Kbit, DLY 10 usec
Full-duplex, 1000Mb/s, media type is 10/100/1000BaseTX
5 minute input rate 41000 bits/sec, 12 packets/sec
 284517 packets input, 39281044 bytes, 0 no buffer
Received 1204 broadcasts (0 multicasts)
 0 runs, 0 giants, 0 throttles
 0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
271033 packets output, 33771201 bytes, 0 underruns
 0 output errors, 0 collisions, 0 interface resets
 0 late collision, 0 deferred
```

That is a healthy port: every error counter zero, full duplex, and the speed matching the media. Learn what this looks like, because you diagnose by difference.

Counter climbing	What it means	Most likely cause
CRC and input errors	Frames arrived damaged	Bad cable, exceeded distance, dirty fibre connector, electrical interference. If collisions is zero, suspect the cable before the duplex
collisions on a full-duplex link	Should be impossible	Duplex mismatch. One end is half, the other full
late collision	A collision after the first 64 bytes	Duplex mismatch, or a segment longer than the standard allows
runs	Frames shorter than 64 bytes	Collisions, or a failing NIC

Counter climbing	What it means	Most likely cause
giants	Frames longer than the MTU	MTU mismatch, or a misconfigured jumbo-frame setting on one side
input errors with CRC zero	Damaged, but not by corruption	Often an MTU or framing problem rather than the cable
output drops	The device could not transmit fast enough	Congestion, not a cable fault. Do not replace the cable

💡 The duplex mismatch, in one paragraph

Auto-negotiation lets two ports agree on speed and duplex. If one end is set to `duplex full` by hand and the other is left on auto, the auto end cannot detect duplex, falls back to half, and you have a mismatch. The half-duplex end listens before transmitting and treats simultaneous transmission as a collision; the full-duplex end transmits whenever it likes. The link stays *up*. It just loses a percentage of every conversation, and the symptom the user reports is “the network is slow”, not “the network is down”.

The fix is to set both ends the same way. The rule of thumb is: leave both on auto, or hard-code both. Never one of each.

➤ Configure — a hard-coded access port, both ends matching

```
interface GigabitEthernet0/1
description Link to SW2 - speed and duplex fixed at both ends
speed 1000
duplex full
no shutdown
```

2.2.1 Fibre and signal levels

Topic 1.1 names *signal levels*. On a fibre link the transceiver reports how much light it is transmitting and receiving, in dBm, and a receive level below the optic’s sensitivity produces exactly the same symptom as a bad copper cable — a link that is up with climbing CRC errors.

🔍 SW1# show interfaces GigabitEthernet0/25 transceiver

Port	Temperature (Celsius)	Optical	
		Voltage (Volts)	Power (dBm)
Gi0/25	33.2	3.31	-28.9

A receive power of -28.9 dBm against a typical sensitivity of about -23 dBm means the link is starved of light. Causes, in the order worth checking: a dirty or unseated connector, a bend or break in the fibre, too many patch panels, a run longer than the optic supports, or a single-mode optic on multimode fibre.

❗ Common Pitfalls

- **Replacing the cable first.** It is the most expensive check in time and the least likely single cause on a link that has been working. Read the counters first: they tell you which of six causes it is.
- **Treating “up/up” as healthy.** A port can be up, passing traffic, and corrupting five per cent of it. The exam will show you exactly this and expect you not to declare it fine.
- **Hard-coding one end.** The commonest way to create a duplex mismatch is to fix one side “to be safe”.

- **Reading raw totals instead of a rate.** Counters accumulate since the last clear. Ten thousand CRC errors over two years is noise; ten thousand since yesterday is a fault. Use `clear counters` and watch.

🔧 Break & Fix — “The file server is slow, but it is not down”.

A user copies a large file and gets a fraction of the expected throughput. Ping succeeds with no loss. You look at the access port:

```
SW1# show interfaces GigabitEthernet0/7 | include duplex|collision|CRC
```

```
Half-duplex, 100Mb/s, media type is 10/100/1000BaseTX
14432 input errors, 118 CRC, 0 frame, 0 overrun, 0 ignored
0 output errors, 89211 collisions, 0 interface resets
2244 late collision, 0 deferred
```

Diagnosis. The port is half duplex at 100 Mbps on gigabit-capable media, with a very large collision count and — decisively — *late* collisions. Late collisions on a modern switched link are the signature of a duplex mismatch: the far end is transmitting full duplex while this end still listens before sending. The small CRC count is a consequence of the collisions, not a separate cable fault.

Fix. Find the server NIC setting. It has almost certainly been hard-coded to full duplex while the switch port was left on auto. Set both ends to auto, or both to `speed 1000 / duplex full`, then `clear counters` and confirm the errors stop climbing. If they do not, *now* suspect the cable.

🔧 Lab 2.1 — Reading a broken link (*Packet Tracer*)

Topology. Two 2960 switches connected by one link; a PC on each.

Task 1. Cable SW1 Gi0/1 to SW2 Gi0/1. Confirm both PCs can ping each other, then run `show interfaces Gi0/1` on both and record the speed, duplex and every error counter while the link is healthy.

Task 2. On SW1 only, set `speed 100` and `duplex half`. Leave SW2 on auto.

Task 3. Generate traffic between the PCs. Record what changes on each side. Which counter moved first?

Task 4. Repeat with SW1 at `duplex full` and SW2 on auto — the classic mismatch. Note that the link stays up.

Task 5. Extension. Without looking at your own configuration, hand the topology to a partner, have them introduce one fault from this chapter, and diagnose it from `show output` alone in under three minutes.

Verification. You should be able to state, from counters only, whether a link is healthy, mismatched, or physically damaged.

❓ Checkpoint — Test Yourself

1. A gigabit link is up. `collisions` is increasing and the interface reports full duplex. What single conclusion can you draw, and why is it certain?
2. A fibre link shows Rx power of -31 dBm against a sensitivity of -23 dBm. Give three causes in the order you would check them.
3. Why is a port that is administratively down easier to deal with than one that is up with rising CRC errors?

 Chapter Summary

- Copper is 100 m, full stop. Multimode reaches a campus; single-mode reaches a city. Exceeding a medium's reach corrupts frames rather than dropping the link.
- Diagnose from counters, not from intuition. CRC without collisions points at the cable; collisions on a full-duplex link point at duplex; late collisions are close to proof of a duplex mismatch.
- Set both ends of a link the same way — both auto, or both fixed.
- “Up/up” does not mean healthy. The dangerous fault is the one that passes most traffic and damages the rest.
- On fibre, read `show interfaces ... transceiver`. Light levels are a diagnosable fault the blueprint names explicitly.

 Review Questions

1. **(MCQ)** Which counter most strongly indicates a duplex mismatch? (a) giants (b) late collision (c) output drops (d) no buffer
2. **(MCQ)** A 10 Gbps link is required over 2 km between two buildings. Which medium? (a) Cat6a (b) multimode OM4 (c) single-mode fibre (d) direct-attach copper
3. **(Short)** Explain why hard-coding duplex on one end of a link and leaving the other on auto produces a mismatch rather than a failure to link.
4. **(Short)** A port shows 8,000 CRC errors and zero collisions. State the two most likely causes and the command you would run next.
5. **(Applied)** You inherit a network where every access port has been hard-coded to `speed 100` and `duplex full`. Describe the risk this creates, and the safest order in which to migrate the estate back to auto-negotiation.

Ethernet, Frames and Switching

EXAM TOPICS 1.1

🎯 Learning Outcomes

By the end of this chapter you will be able to:

- **Describe** the Ethernet frame and the purpose of each field you will actually meet on a device.
- **Explain** how a switch learns, forwards, floods and ages MAC addresses — the algorithm behind every switching chapter that follows.
- **Read** the MAC address table and use it to prove where a device is physically connected.
- **Predict** what a switch does with a unicast, broadcast or unknown destination.

🔗 Before You Start

Chapter 1 for frames and broadcast domains; Chapter 2 for the link underneath.

📖 Key Terms in This Chapter

Ethernet • MAC address • OUI • unicast • broadcast • multicast • CAM table • MAC address table • learning • flooding • filtering • ageing • unknown unicast • MTU • FCS

3.1 The frame

📄 The Ethernet frame, field by field

- **Preamble** — clock synchronisation. You will never see it.
- **Destination MAC** (6 bytes) — who this frame is for. The switch reads this to decide where to send it.
- **Source MAC** (6 bytes) — who sent it. The switch reads this to *learn*.
- **Type/Length** (2 bytes) — what is inside. 0x0800 is IPv4, 0x86DD is IPv6, 0x8100 means a VLAN tag follows (Chapter 10).
- **Payload** — 46 to 1500 bytes. Below 46 the frame is padded; this is why a frame under 64 bytes total is a **runt**.
- **FCS** (4 bytes) — a checksum. If it fails, the frame is counted as a CRC error and discarded (Chapter 2).

A **MAC address** is 48 bits, written as twelve hex digits. The first half is the **OUI**, assigned to the manufacturer; the second half is assigned by that manufacturer. Cisco writes them as 00d0.58a1.0201; almost everyone else writes 00:d0:58:a1:02:01. They are the same address, and you should be able to read both.

💡 Three kinds of destination

- **Unicast** — one specific device. The low-order bit of the first byte is 0.
- **Broadcast** — ffff.ffff.ffff, every device in the VLAN. This is what defines a broadcast domain.

- **Multicast** — a group. The low-order bit of the first byte is 1. OSPF (Chapter 17), HSRP and VRRP (Chapter 19) all use multicast.

3.2 How a switch actually works

A switch runs one small algorithm, and everything in Part II is a modification of it.

💡 Learn, then forward or flood

For every frame that arrives on a port:

1. **Learn.** Record the *source* MAC address against the port it arrived on, in this VLAN. Reset the ageing timer if it is already there.
2. **Decide on the destination.**
 - Known unicast → **forward** out that one port.
 - Unknown unicast → **flood** out every other port in the VLAN.
 - Broadcast or multicast → **flood**.
 - Destination is on the same port it arrived on → **filter**, that is, discard it.

A switch never modifies the addresses in a frame it forwards, and it never consults an IP address. Both of those are the router's job.

⚠️ The consequence that matters: unknown unicast flooding

A switch floods a unicast frame when it does not know the destination. That is not a fault — it is how the network keeps working while the table is learned. But it means a full or wiped MAC table causes traffic to be sprayed to every port, which is both a performance problem and a security one. It is also the mechanism a MAC flooding attack abuses, which is why Chapter 25 exists and why topic 4.7 names port security.

Entries age out after five minutes of silence by default. A device that has been quiet longer than that has to be re-learned, and the first frame to it is flooded.

🔍 SW1# show mac address-table

Mac Address Table			
Vlan	Mac Address	Type	Ports
1	00d0.58a1.0201	DYNAMIC	Gi0/1
1	0060.5c2b.19a4	DYNAMIC	Gi0/2
10	000a.f36e.7712	DYNAMIC	Gi0/5
10	0002.1751.4c33	STATIC	Gi0/6
20	00e0.b05f.2210	DYNAMIC	Gi0/8
Total Mac Addresses for this criterion: 5			

This table is the most useful single output on a switch. It answers “where is this device physically plugged in?”, and it does so by evidence rather than by documentation — which is exactly what topic 2.3 asks you to do when it says *validate the accuracy of network documentation*.

➤ Configure — a static MAC entry, and a shorter ageing time

```
! Pin a known device to a known port. Rarely needed, but it survives
! ageing and cannot be moved by a spoofed frame.
mac address-table static 0002.1751.4c33 vlan 10 interface GigabitEthernet0/6
!
! Age entries out faster than the 300-second default.
mac address-table aging-time 120
```

Worked Example: Following a frame across two switches

PC-A (00d0.58a1.0201) on SW1 Gi0/1 sends to PC-B (00e0.b05f.2210) on SW2 Gi0/3. Both switches have empty tables.

1. The frame arrives at SW1 Gi0/1. SW1 *learns* 00d0.58a1.0201 on Gi0/1.
2. SW1 does not know PC-B, so it *floods* — out its uplink and every other access port.
3. SW2 receives it on its uplink and learns 00d0.58a1.0201 on *that uplink port*, not on Gi0/1. A switch only ever knows the direction, not the distance.
4. SW2 does not know PC-B either, so it floods. PC-B receives it.
5. PC-B replies. Now SW2 learns PC-B on Gi0/3, and because it already knows PC-A, it *forwards* rather than flooding.
6. SW1 learns PC-B on its uplink. From here on the conversation is unicast in both directions. Two floods to start a conversation, then none. That is the whole design.

Common Pitfalls

- **Expecting a switch to know a distance.** The table says which port, never how many hops. Five hundred devices can appear behind one uplink port and that is entirely normal.
- **Reading a stale table.** An entry can be up to five minutes old. If you have just moved a cable, the table may still show the old port.
- **Confusing the MAC table with the ARP table.** The MAC table maps MAC → port and lives on a switch. The ARP table maps IP → MAC and lives on any device with an IP address. The exam will test that you know which to look at.
- **Forgetting the VLAN column.** The same MAC address can legitimately appear in two VLANs. Learning is always per VLAN.

Break & Fix — Two ports, one MAC address, and a flapping log.

Users on VLAN 10 report intermittent drops. The log shows:

```
SW1# show logging | include MACFLAP
```

```
%SW_MATM-4-MACFLAP_NOTIF: Host 000a.f36e.7712 in vlan 10 is flapping
between port Gi0/5 and port Gi0/12
```

Diagnosis. The switch is learning the same source MAC on two ports in quick succession. A device cannot be in two places, so one of three things is true: there is a **physical loop** joining those ports (the same frame is arriving by two paths — see Chapter 12), a device is **spoofing** that address, or two devices have been given the **same MAC** by a bad clone or a virtualisation misconfiguration.

Fix. Check spanning tree first, because a loop is both the most likely cause and the most damaging: `show spanning-tree vlan 10` and look for a port that should be blocking and is not. If the topology is loop-free, shut Gi0/12, confirm the flapping stops, and trace what is attached to it.

Lab 3.1 — Watching a switch learn (*Packet Tracer*)

Topology. Two switches, uplinked; two PCs on SW1, one on SW2. All in VLAN 1 for now.

Task 1. Power everything on and immediately run `show mac address-table` on both switches. Record what is there before any traffic.

Task 2. Ping from PC-A to PC-C. Re-run the command on both switches and record which addresses each has learned, and against which port.

- Task 3.** Explain, from your own capture, why SW2 shows PC-A against its uplink rather than against a host port.
- Task 4.** Run `clear mac address-table dynamic`, then ping again and watch the table rebuild.
- Task 5.** Set `mac address-table aging-time 15`. Leave a PC idle for thirty seconds and confirm its entry disappears.
- Task 6. Extension.** Add a second cable between the two switches to create a loop. Observe what happens to the MAC table and the log before spanning tree settles — then read Chapter 12 to find out why the network survived.

Checkpoint — Test Yourself

1. A switch receives a frame whose destination MAC is already in its table, against the same port the frame arrived on. What does it do, and what is that behaviour called?
2. Why does the first ping between two devices often show a longer round trip than the ones after it?
3. You need to prove which wall port a particular laptop is using. Which table do you consult, on which device, and what do you need to know first?

Chapter Summary

- A switch learns from the *source* address and decides on the *destination* address. Everything else follows.
- Known unicast is forwarded, unknown unicast and broadcast are flooded, and a frame is never sent back out of the port it came in on.
- The MAC address table maps MAC to port, per VLAN, and ages entries out after five minutes by default.
- MAC table is not ARP table. One lives on a switch and maps to ports; the other maps IP to MAC and lives on anything with an IP address.
- A MAC address flapping between two ports means a loop, a spoof, or a duplicate — and you check for the loop first.

Review Questions

1. **(MCQ)** A switch receives a frame destined for `ffff.ffff.ffff`. What does it do? (a) forwards it to the router (b) floods it out all ports in the VLAN except the ingress port (c) drops it (d) sends it to the CPU
2. **(MCQ)** Which field does a switch use to populate its MAC address table? (a) destination MAC (b) source MAC (c) source IP (d) Type/Length
3. **(Short)** Explain why a switch shows hundreds of MAC addresses against a single uplink port, and why this is not a fault.
4. **(Short)** Give two distinct causes of a MAC flap message and say which you would investigate first, with a reason.
5. **(Applied)** A security auditor asks you to prove that no unauthorised device is connected to the finance VLAN. Describe, with commands, how you would produce that evidence and what its limitations are.

IPv4 Addressing and Subnetting

EXAM TOPICS **1.3**

🎯 Learning Outcomes

By the end of this chapter you will be able to:

- **Convert** between dotted decimal, binary and prefix length fast enough to do it in an exam without a calculator.
- **Compute** the network address, broadcast address, usable range and host count of any subnet, in under thirty seconds.
- **Subnet** a given block to meet a stated requirement, and apply variable-length masks without overlapping.
- **Diagnose** the four addressing faults that account for almost every broken host: wrong mask, wrong gateway, address outside the subnet, and duplicate address.

🔗 Before You Start

Chapter 1 and Chapter 3. You need to be comfortable that a switch works on MAC addresses and a router works on IP addresses.

📖 Key Terms in This Chapter

dotted decimal • prefix length • subnet mask • network address • broadcast address • usable range • block size • magic number • RFC 1918 • private address • public address • VLSM • default gateway • APIPA

🎓 This topic is worth more time than any other in Part I

Topic 1.3 says *troubleshoot* IPv4 addressing, assignment and subnetting, “public and private”. You will be given a host that cannot reach something and expected to spot that its mask is wrong. Every second you spend computing a subnet boundary is a second not spent reasoning, so the arithmetic has to be automatic. Learn the method in §4.2 until it is boring.

4.1 The address and the mask

An IPv4 address is 32 bits, written as four decimal bytes. It carries two pieces of information at once, and the mask is what separates them.

📖 Network part and host part

The **subnet mask** marks, with 1 bits, which leading bits of the address identify the **network**. The remaining 0 bits identify the **host** within it. Written as a count of 1 bits, the mask is a **prefix length**: 255.255.255.0 and /24 are the same statement.

Two addresses are on the same subnet if and only if their network parts are identical. That single sentence is the source of most addressing faults: a host with the wrong mask computes a different network part, decides its neighbour is remote, and sends the traffic to a gateway that may not want it.

Prefix	Mask	Block	Usable	Typical use
/24	255.255.255.0	256	254	A standard user VLAN
/25	255.255.255.128	128	126	Half a /24
/26	255.255.255.192	64	62	A small office
/27	255.255.255.224	32	30	A rack or a server segment
/28	255.255.255.240	16	14	A tiny segment
/29	255.255.255.248	8	6	Four routers and room to grow
/30	255.255.255.252	4	2	A point-to-point link between routers
/31	255.255.255.254	2	2	A point-to-point link, no waste (RFC 3021)
/32	255.255.255.255	1	1	A host route or a loopback

Usable is block size minus two, because the first address in a block is the network address and the last is the broadcast address. The exception is /31, where both are usable because a point-to-point link has no need for a broadcast.

Private addresses — RFC 1918

Three ranges are reserved for internal use and are never routed on the internet. The exam expects you to recognise them instantly.

- 10.0.0.0/8 — one very large block
- 172.16.0.0/12 — that is 172.16.0.0 to 172.31.255.255, and the upper bound catches people out
- 192.168.0.0/16

Also worth knowing: 169.254.0.0/16 is **APIPA**, which a host assigns itself when DHCP fails. A host with a 169.254 address has not been given one — see Chapter 20.

4.2 The method: block size, or “magic number”

You will not have time to write out binary. This method needs no binary at all.

Four steps, every time

1. **Find the interesting octet** — the one where the mask is neither 255 nor 0.
2. **Block size** = 256 – (the mask value in that octet). This is the magic number.
3. **Count in blocks** from zero in that octet until you pass the address. The multiple you last passed is the **network address**.
4. **Broadcast** = next network address – 1. Usable range is everything between the two, exclusive.

Worked Example: Where does 172.16.83.190/26 live?

1. Mask for /26 is 255.255.255.192. The interesting octet is the fourth.
2. Block size = 256 – 192 = 64.
3. Count in 64s in the fourth octet: 0, 64, 128, **192**. Our host is 190, which sits after 128 and before 192. Network address is 172.16.83.128.

- Next network is 172.16.83.192, so broadcast is 172.16.83.191. Usable range is 172.16.83.129 to 172.16.83.190.

So .190 is the *last usable address* in its subnet — a perfectly valid host address, and a favourite exam distractor because it looks like a broadcast.

Worked Example: The same method in the third octet: 10.44.150.7/21

- /21 is 255.255.248.0. Interesting octet is the third.
- Block size = $256 - 248 = 8$.
- Count in 8s in the third octet: 0, 8, ..., 144, **152**. Our 150 sits after 144. Network is 10.44.144.0.
- Next network is 10.44.152.0, so broadcast is 10.44.151.255. Usable is 10.44.144.1 to 10.44.151.254 — 2046 hosts.

Note that the fourth octet plays no part in finding the boundary. Only the interesting octet does.

4.3 Subnetting a block to a requirement

Worked Example: Split 192.168.20.0/24 for four sites of 50, 25, 10 and 2 hosts

Work *largest first*, or you will strand addresses.

- 50 hosts needs 62 usable → /26, block 64. 192.168.20.0/26 — usable .1–.62.
- 25 hosts needs 30 usable → /27, block 32. Next free address is .64: 192.168.20.64/27 — usable .65–.94.
- 10 hosts needs 14 usable → /28, block 16. Next free is .96: 192.168.20.96/28 — usable .97–.110.
- 2 hosts (a router link) → /30, block 4. Next free is .112: 192.168.20.112/30 — usable .113–.114.

Everything from .116 upwards is still free. This is **VLSM**: different masks inside one block, sized to the requirement.

The two rules VLSM must obey

- A subnet must start on a multiple of its own block size.** A /27 can start at .64 or .96, never at .70.
- Subnets must not overlap.** If you allocate smallest-first you will eventually place a large subnet where it does not fit, and have to start again. Largest first avoids this.

4.4 Configuring and verifying an address

Configure — an interface address and a default gateway

```
! On a router interface -- the mask is written in full.
interface GigabitEthernet0/0
  description User VLAN 10 gateway
  ip address 192.168.20.1 255.255.255.192
  no shutdown
!
! On a layer 2 switch, the management address lives on an SVI,
! and the switch needs a default gateway because it does not route.
interface Vlan10
```

```
ip address 192.168.20.2 255.255.255.192
no shutdown
!
ip default-gateway 192.168.20.1
```

 R1# show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	192.168.20.1	YES	manual	up	up
GigabitEthernet0/1	unassigned	YES	unset	administratively down	down
Vlan1	unassigned	YES	unset	down	down

Reading the two status columns

The pair of columns is the single most useful diagnostic on a router, and the combinations mean different things.

- **up / up** — working.
- **up / down** — the cable and electrical link are fine, but the protocol is not. Usually an encapsulation or keepalive mismatch, or the far end is not configured.
- **down / down** — a physical problem. Cable, transceiver, or the other end is shut. Go back to Chapter 2.
- **administratively down / down** — somebody typed `shutdown`. The fix is `no shutdown`.

Common Pitfalls

- **Forgetting `no shutdown`.** Router interfaces are shut by default. Switch ports are not. This asymmetry catches everyone once.
- **Assuming 172.16.0.0/12 stops at 172.16.** It runs to 172.31.255.255. 172.32.0.0 is public.
- **Giving the gateway an address outside the subnet.** A host will accept the configuration and then fail silently for everything off-net.
- **Using the network or broadcast address on a host.** .0 and .255 are only network and broadcast in a /24. In a /26 the boundaries move, and 192.168.20.64 is a network address even though it does not end in zero.

Break & Fix — “I can reach my own floor but nothing else”.

A host is configured 192.168.20.100 mask 255.255.255.192, gateway 192.168.20.1. It can ping 192.168.20.90 but not its gateway and nothing beyond.

Diagnosis. Work the mask. /26 gives block size 64, so the boundaries are .0, .64, .128, .192. The host at .100 is in the 192.168.20.64/26 subnet, usable .65–.126. Its gateway at .1 is in the *first* subnet, 192.168.20.0/26. The host and its gateway are on different subnets, so the host cannot ARP for it and has no way off its own segment. It can still reach .90 because that address is in the same /26.

Fix. Either move the host into the subnet its gateway serves, or — far more likely what was intended — correct the gateway to 192.168.20.65, the first usable address of the subnet the host is actually in. Then confirm with a ping to the gateway before testing anything remote.

Lab 4.1 — Subnet, configure, then break it (*Packet Tracer*)

Topology. One router, two switches, four PCs — two per VLAN.

Task 1. Subnet 172.16.5.0/24 into two subnets of at least 100 hosts each. Write down, before touching a device: network address, first and last usable address, and broadcast address for both.

- Task 2.** Configure the router with a gateway address in each subnet and the PCs with addresses from your plan. Verify with `show ip interface brief` and a ping within each subnet.
- Task 3.** Prove routing works between the two subnets by pinging across.
- Task 4.** Now change one PC's mask from your planned value to 255.255.255.0. Predict what will break *before* you test, then test it. Explain the result in terms of the network part.
- Task 5.** Change it back, then set one PC's gateway to an address in the other subnet. Predict and test again.
- Task 6. Extension.** Re-plan the same /24 using VLSM for four segments of 100, 50, 20 and 2 hosts, and configure it without overlaps.

Checkpoint — Test Yourself

- For 10.19.200.37/27: give the network address, broadcast address, and first and last usable address.
- A host is 192.168.1.130/25 and its gateway is 192.168.1.126. Will it work? Show your reasoning.
- How many usable addresses does a /30 provide, and why is a /31 sometimes used instead on a router-to-router link?

Chapter Summary

- Address plus mask equals two facts: which network, and which host. Devices are on the same subnet only if their network parts match.
- Block size = 256 – mask octet. Count in blocks. That method answers every subnetting question you will be asked.
- Usable = block – 2, except /31 and /32.
- Private ranges are 10/8, 172.16/12 (to 172.31) and 192.168/16. A 169.254 address means DHCP failed.
- Plan VLSM largest first, and start every subnet on a multiple of its own block size.
- Four faults cover most cases: wrong mask, wrong gateway, address outside the subnet, duplicate address.

Review Questions

- (MCQ)** A host is addressed 192.168.90.200/28. What is the broadcast address of its subnet? (a) 192.168.90.207 (b) 192.168.90.255 (c) 192.168.90.191 (d) 192.168.90.215
- (MCQ)** Which is a valid host address in 10.0.8.0/22? (a) 10.0.8.0 (b) 10.0.11.255 (c) 10.0.11.254 (d) 10.0.12.1
- (Short)** Explain why a host with the correct address but a mask that is too short may still reach some destinations and not others.
- (Short)** A host has address 169.254.14.9. What has happened, and what would you check first?
- (Applied)** You are given 192.168.40.0/24 and asked for six subnets: 60, 60, 28, 12, 12 and 2 hosts. Produce a VLSM plan with no overlaps, showing network address, mask and usable range for each, and state how much of the block remains free.

IPv6 Addressing

EXAM TOPICS **1.4**

🎯 Learning Outcomes

By the end of this chapter you will be able to:

- **Abbreviate** and expand an IPv6 address correctly, applying both compression rules without ambiguity.
- **Identify** the address types the exam names: global unicast, link local, unique local, multicast, and the solicited-node address.
- **Construct** an interface identifier by modified EUI-64, which the blueprint calls out explicitly.
- **Size** a prefix, and **diagnose** the assignment faults that stop an IPv6 host communicating.

🔗 Before You Start

Chapter 4 — the ideas of network part, host part and prefix length carry over unchanged. Chapter 3 for MAC addresses, which EUI-64 is built from.

📖 Key Terms in This Chapter

global unicast • link local • unique local • multicast • solicited-node • anycast • modified EUI-64 • SLAAC • router advertisement • neighbour discovery • dual stack • prefix

🎓 What v2.0 asks, and what it dropped

Topic 1.4 says *troubleshoot* IPv6 address configuration, assignment and prefix sizing, and narrows the scope to “unicast and modified EUI 64”. The previous blueprint asked you to compare every address type in detail; this one does not. Concentrate on getting an address onto an interface, understanding where the link-local one came from, and spotting a wrong prefix.

5.1 Writing the address

An IPv6 address is 128 bits, written as eight groups of four hex digits separated by colons. Two rules compress it.

💡 The two compression rules

1. **Drop leading zeros** in any group. 0db8 becomes db8; 0000 becomes 0. Trailing zeros are never dropped.
2. **Replace one run of all-zero groups with ::**. You may do this **once only**, because a second :: would make the address ambiguous — there would be no way to know how many zeros each stood for.

📊 Worked Example: Compressing an address

Start with the full form:

2001:0db8:0000:0000:0000:ff00:0042:8329

1. Drop leading zeros in each group: 2001:db8:0:0:0:ff00:42:8329

2. Replace the longest run of zero groups with `::`: `2001:db8::ff00:42:8329`

Expanding is the same in reverse. Given `2001:db8::1`, count the groups present — two before the `::` and one after, so three — and the `::` must therefore stand for five zero groups: `2001:0db8:0000:0000:0000:0000:0000:0001`.

5.2 The types you must recognise

Type	Prefix	What it is for
Global unicast	<code>2000::/3</code>	Routable on the internet. In practice you meet <code>2001:db8::/32</code> , which is reserved for documentation and used throughout this book
Link local	<code>fe80::/10</code>	Valid on one link only, never routed. Every IPv6 interface has one automatically, and routing protocols use it as the next hop
Unique local	<code>fc00::/7</code>	The rough equivalent of RFC 1918 private space. In practice <code>fd00::/8</code>
Multicast	<code>ff00::/8</code>	Groups. <code>ff02::1</code> is all nodes on the link, <code>ff02::2</code> is all routers, <code>ff02::5</code> is OSPFv3
Solicited-node	<code>ff02::1:ff00:0/104</code>	Used by neighbour discovery in place of ARP. Formed from the last 24 bits of the unicast address
Loopback	<code>::1</code>	The equivalent of <code>127.0.0.1</code>
Unspecified	<code>::</code>	“No address yet”, used as a source during autoconfiguration

⚠ There is no broadcast in IPv6

IPv6 has no broadcast address at all. Where IPv4 would broadcast, IPv6 uses a multicast group — usually `ff02::1`, all nodes. This is why ARP does not exist in IPv6: its job is done by neighbour discovery using the solicited-node multicast address, which reaches only the handful of hosts whose addresses end the same way rather than every device on the segment.

5.3 Modified EUI-64

The blueprint names this explicitly, so it is examinable arithmetic.

💡 Turning a 48-bit MAC into a 64-bit interface ID

1. Split the MAC in half.
2. Insert `fffe` in the middle.
3. **Flip the seventh bit** of the first byte — the universal/local bit. In hex this means adding or subtracting 2 from the second hex digit.

📊 Worked Example: EUI-64 from `00d0.58a1.0201`

1. Split: `00d0.58` and `a1.0201`.
2. Insert `fffe`: `00d0:58ff:fea1:0201`.
3. Flip the seventh bit of `00`. In binary `00` is `0000 0000`; flipping the seventh bit gives `0000 0010`, which is `02`.

4. Interface ID is 02d0:58ff:fea1:0201. With a prefix of 2001:db8:acad:10::/64 the full address is 2001:db8:acad:10:2d0:58ff:fea1:201.

The quick check: an address containing ff:fe in the middle of its host portion was almost certainly built by EUI-64.

5.4 Configuring IPv6

>_ Configure — IPv6 on a router interface, three ways

```
! Routing must be enabled globally first -- this is the step most often
! forgotten, and without it the router will not forward or advertise.
ipv6 unicast-routing
!
interface GigabitEthernet0/0
description User VLAN 10 gateway
! 1. A fully specified address.
ipv6 address 2001:db8:acad:10::1/64
! 2. Or let the device build the host part by EUI-64.
ipv6 address 2001:db8:acad:20::/64 eui-64
! 3. A link-local address is created automatically, but you can pin it
! to something memorable -- worth doing, because routing protocols
! show it as the next hop.
ipv6 address fe80::1 link-local
no shutdown
```

Q R1# show ipv6 interface brief

```
GigabitEthernet0/0    [up/up]
FE80::1
2001:DB8:ACAD:10::1
2001:DB8:ACAD:20:2D0:58FF:FEA1:201
GigabitEthernet0/1    [administratively down/down]
unassigned
```

Note that one interface holds several addresses at once, and that this is normal rather than a fault. An IPv6 interface essentially always has a link-local address plus whatever global addresses it has been given.

💡 How a host gets an address: SLAAC versus DHCPv6

A host sends a router solicitation to ff02::2; a router replies with a **router advertisement** carrying the prefix. Then either:

- **SLAAC** — the host builds its own address from the advertised prefix plus an interface ID it generates. No server involved.
- **DHCPv6** — the advertisement tells the host to ask a server, which supplies the address and other options.

A host with only a fe80:: address and no global one has heard no router advertisement. That is the IPv6 equivalent of an APIPA address, and it points at the router, not the host.

⚠ Common Pitfalls

- **Omitting ipv6 unicast-routing.** Interfaces come up and addresses look right, but the router neither forwards nor advertises.
- **Using :: twice.** It is not merely discouraged, it is invalid, because the address can no longer be expanded uniquely.

- **Dropping trailing zeros.** 2001:db8:1000:: cannot be shortened to 2001:db8:1::.. Only *leading* zeros go.
- **Assuming a prefix other than /64 for a LAN.** SLAAC requires a 64-bit interface ID, so a LAN prefix longer than /64 breaks autoconfiguration. Point-to-point links are the usual exception.

Break & Fix — Dual-stack host reaches IPv4 destinations only.

A workstation has a working IPv4 address and, for IPv6, only fe80::2d0:58ff:fea1:201. It cannot reach any IPv6 destination, on or off the segment.

Diagnosis. A link-local address is generated by the host itself and proves nothing except that the interface is up. The absence of a global address means no router advertisement is being received. Either no router is configured for IPv6 on that VLAN, or `ipv6 unicast-routing` is missing, or the router interface has no global prefix to advertise.

Fix. On the gateway, confirm `ipv6 unicast-routing` is present in the running configuration, then check `show ipv6 interface GigabitEthernet0/0` shows a global address and that router advertisements are not suppressed. Bring the prefix back and the host will build its own address within seconds — no change is needed on the host at all.

Lab 5.1 — Dual-stack a small network (*Packet Tracer*)

Topology. One router, one switch, two PCs, addressed for both protocols.

- Task 1.** Enable `ipv6 unicast-routing` and give Gi0/0 the address 2001:db8:acad:10::1/64 and link-local fe80::1.
- Task 2.** Set both PCs to obtain IPv6 automatically. Record the addresses they build and confirm they contain ff:fe.
- Task 3.** Compute by hand what each PC's EUI-64 address should be from its MAC, and check your answer against the device.
- Task 4.** Ping between the PCs using IPv6, then ping the router's link-local address from a PC. Note that the link-local ping needs no global addressing at all.
- Task 5.** Remove `ipv6 unicast-routing` and observe what happens to the PCs after their addresses expire.
- Task 6. Extension.** Configure a second VLAN with prefix 2001:db8:acad:20::/64 and route between the two.

Checkpoint — Test Yourself

1. Compress 2001:0db8:0000:0012:0000:0000:0000:00ab fully.
2. A host shows only an fe80:: address. What is missing, and on which device would you look?
3. Build the modified EUI-64 interface identifier for MAC aabb.cc00.0100.

Chapter Summary

- Drop leading zeros; use :: once, for the longest zero run.
- 2000::/3 global, fe80::/10 link local, fd00::/8 unique local, ff00::/8 multicast. There is no broadcast.
- EUI-64: split the MAC, insert fffe, flip the seventh bit.
- `ipv6 unicast-routing` is a separate, easily forgotten step.
- LANs are /64 because SLAAC needs a 64-bit interface ID.

- Only a link-local address means no router advertisement was heard — investigate the router, not the host.

Review Questions

1. **(MCQ)** Which address is a valid compression of 2001:0db8:0000:0000:1000:0000:0000:0001?
(a) 2001:db8::1000::1 (b) 2001:db8:0:0:1::1 (c) 2001:db8::1000:0:0:1 (d) 2001:db8:1000::1
2. **(MCQ)** Which multicast group do all IPv6 routers on a link join? (a) ff02::1 (b) ff02::2
(c) ff02::5 (d) ff02::9
3. **(Short)** Explain why :: may appear only once in an address.
4. **(Short)** A LAN is configured with a /112 prefix and hosts are set to autoconfigure. What will happen, and why?
5. **(Applied)** You are dual-stacking an existing IPv4 network. Describe the order in which you would enable IPv6 on the router, verify each step, and what evidence you would collect before declaring the segment working.

TCP, UDP and the Diagnostic Toolkit

EXAM TOPICS **2.4**

🎯 Learning Outcomes

By the end of this chapter you will be able to:

- **Explain** what a port number identifies, and recognise the well-known ports that appear in access lists and in fault reports.
- **Choose** correctly between TCP and UDP behaviour when reasoning about a symptom.
- **Use** ping, extended ping and traceroute deliberately, varying source, size and count to test a specific hypothesis.
- **Read** packet capture output well enough to say which direction failed.

🔗 Before You Start

Chapter 4 for addressing, Chapter 3 for what a switch does with a frame.

📖 Key Terms in This Chapter

port • socket • TCP • UDP • three-way handshake • acknowledgement • retransmission • well-known port • ephemeral port • ICMP • echo request • TTL • extended ping • traceroute

🎓 The toolkit is the exam

Topic **2.4** names the tools by name: “show commands (including show logs), ping, extended ping, trace route, and packet capture output”. It is one of the eight *troubleshoot* topics, and the tools in it are how you will answer the other seven. Learn them as instruments, not as trivia.

6.1 Ports, and what they identify

An IP address identifies a host. A **port number** identifies which program on that host. Together they form a **socket**, and a conversation is defined by four things: source address, source port, destination address, destination port.

Port	Service	Protocol	Why it matters here
22	SSH	TCP	Device management. Chapter 23
23	Telnet	TCP	Device management, unencrypted. Disable it
53	DNS	UDP, TCP	Name resolution. Chapter 21
67, 68	DHCP	UDP	Address assignment. Chapter 20
69	TFTP	UDP	Legacy file transfer. Replaced by SFTP/SCP in v2.0
80, 443	HTTP, HTTPS	TCP	The traffic users notice
161, 162	SNMP	UDP	Monitoring and traps. Chapter 30
514	Syslog	UDP	Log delivery. Chapter 30

Client programs use a high-numbered **ephemeral** port as their source. This is why a return packet is not simply “port 80” — it is addressed to whatever ephemeral port the client opened, and why stateless access lists need care (Chapter 24).

💡 TCP or UDP, and what each failure looks like

TCP establishes a connection with a three-way handshake — SYN, SYN-ACK, ACK — numbers every byte, acknowledges what arrives and retransmits what does not. It costs latency and gains reliability.

UDP sends and forgets. No handshake, no acknowledgement, no retransmission. It is right for anything where a late packet is worse than a lost one, and for short exchanges where a handshake would double the cost — which is why DNS and DHCP use it.

The diagnostic value is in how they fail. A blocked TCP port produces a *connection timeout or refusal* — something definite. A blocked UDP port often produces *nothing at all*, because there was never going to be an acknowledgement. “It just hangs” points at UDP or at a one-way path; “connection refused” means something answered and said no.

6.2 Ping, and how to ping deliberately

Ping sends an ICMP echo request and waits for a reply. It proves that a path exists in *both* directions — a point worth dwelling on, because a successful ping rules out far more than a failed one confirms.

Q R1# ping 192.168.20.10

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.20.10, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/4 ms
```

Symbol	Meaning
!	Reply received. The path works in both directions
.	Timed out. Either no path, no return path, or something is dropping silently
U	Destination unreachable — a router explicitly told you it has no route. This is far more informative than a timeout
Q	Source quench: congestion
M	Could not fragment. An MTU problem

💡 The extended ping is the one that finds faults

A plain ping from a router sources the packet from the exit interface. That is often not the address you care about. **Extended ping** lets you set the source, and this single option answers the commonest routing question: “does the far end have a route *back* to this subnet?” Run ping with no arguments to enter the extended dialogue.

Q R1# ping

```
Protocol [ip]:
Target IP address: 192.168.30.10
Repeat count [5]: 100
Datagram size [100]: 1500
Timeout in seconds [2]:
Extended commands [n]: y
Source address or interface: 192.168.20.1
Type of service [0]:
```

```
Set DF bit in IP header? [no]: yes
Sweep range of sizes [n]:
Sending 100, 1500-byte ICMP Echos to 192.168.30.10, timeout is 2 seconds:
Packet sent with a source address of 192.168.20.1
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
Success rate is 100 percent (100/100), round-trip min/avg/max = 1/3/12 ms
```

- **Source address** — test the return path for a specific subnet.
- **Repeat count** — 100 pings exposes intermittent loss that five will miss.
- **Datagram size with DF bit set** — finds MTU problems. If 1500 fails and 1400 succeeds, something on the path cannot carry a full-size frame.

6.3 Traceroute

Traceroute sends packets with deliberately small TTL values. The first router decrements TTL to zero, discards the packet and reports back; the second does the same for TTL 2, and so on. The result is a list of hops.

```
Q R1# traceroute 192.168.30.10
```

```
Type escape sequence to abort.
Tracing the route to 192.168.30.10

 0  10.0.12.2  4 msec  1 msec  1 msec
 1  10.0.23.3  8 msec  5 msec  4 msec
 2  * * *
 3  * * *
```

! Where a traceroute stops is not always where the fault is

Asterisks mean no reply from that hop. That can mean the fault is there — or that the hop is configured not to reply, or that the *return* path from that hop is broken while the forward path is fine. A traceroute that dies at hop 3 narrows the search to hop 3 onwards; it does not convict hop 3. Confirm with an extended ping sourced from an address the far end should know.

6.4 Reading a capture

```
Q SW1# show capture BUF
```

```
 1  0.000000  192.168.20.10 -> 192.168.30.10  ICMP Echo request  id=1 seq=1
 2  2.001113  192.168.20.10 -> 192.168.30.10  ICMP Echo request  id=1 seq=2
 3  4.002240  192.168.20.10 -> 192.168.30.10  ICMP Echo request  id=1 seq=3
```

Three requests, no replies. That is a one-way path: the packets are leaving and arriving, but nothing is coming back. The fault is on the return path or at the destination host, and it is emphatically *not* the local switch — which is exactly the conclusion the capture is there to let you draw.

! Common Pitfalls

- **Treating a failed ping as proof of a broken forward path.** Ping needs both directions. Half the time the forward path is fine.
- **Pinging from the wrong source.** A router pinging from its exit interface tests a path the users never take.
- **Concluding from five packets.** Intermittent loss needs a hundred.
- **Forgetting that ICMP may be filtered.** Plenty of hosts and firewalls drop echo requests by policy. “It does not ping” is not the same as “it is down”.

Break & Fix — Web server reachable by IP, refuses connections by name and port.

Users cannot load an internal site. From a workstation, `ping 10.20.5.40` succeeds with 0% loss. Browsing to the address times out.

Diagnosis. Ping proves layer 3 connectivity in both directions, so addressing, masks, routing and the physical path are all sound. The failure is above layer 3 — either the service is not listening, or something is filtering TCP 443 specifically. A timeout rather than a refusal points at a filter: a closed port on a live host normally answers with a reset.

Fix. Test the port rather than the host. From a router on the path, an extended ping cannot help — it is ICMP only — so use `telnet 10.20.5.40 443`: a blank screen means the port is open, a refusal means closed, and a hang means filtered. Then walk the access lists on the path (Chapter 24). Do not touch addressing; the ping already cleared it.

Lab 6.1 — Diagnose with intent (*Packet Tracer*)

Topology. Two routers back to back, one LAN each, one PC per LAN.

Task 1. Address everything and confirm end-to-end ping between the PCs.

Task 2. From R1, run a plain ping to the far PC, then an extended ping sourced from the LAN interface. Explain why the second is the more useful test.

Task 3. Remove the return route on R2. Predict what a ping from PC-A will show, and what a traceroute will show, *before* testing.

Task 4. Restore the route. Now set the far LAN interface MTU to 1400, and use an extended ping with size 1500 and the DF bit set. Record the result.

Task 5. Run a hundred-packet ping and record whether any loss appears.

Task 6. Extension. Have a partner break one thing — an address, a mask, a route or an interface — and identify which, using only ping, extended ping and traceroute, in under five minutes.

Checkpoint — Test Yourself

1. A ping returns U.U.U. What has happened, and why is that more useful than five dots?
2. Why does an extended ping with a specified source often succeed when a plain ping from the same router fails?
3. A traceroute shows three hops then asterisks. List three different explanations.

Chapter Summary

- A port identifies the program; address plus port is a socket.
- TCP handshakes, acknowledges and retransmits; UDP does none of it. Their failure modes differ, and the difference is diagnostic.
- Ping tests both directions at once. A success rules out a great deal; a failure convicts nothing on its own.
- Extended ping sets the source, size and DF bit — it is the tool that finds return-path and MTU faults.
- Traceroute narrows the search; it does not name the culprit.
- A capture showing requests and no replies is a one-way path.

Review Questions

1. (MCQ) Which pair of ping output characters indicates that a router explicitly reported it had no route? (a) (b) UUUUU (c) !!!!! (d) MMMMM

2. **(MCQ)** Which protocol would you expect a DNS query to use by default? (a) TCP 53 (b) UDP 53 (c) UDP 67 (d) TCP 22
3. **(Short)** Explain why setting the source address on a ping is the single most useful extended option when troubleshooting routing.
4. **(Short)** A service “hangs” rather than being refused. What does that suggest, and what would you check?
5. **(Applied)** A user reports a file transfer that succeeds for small files and fails for large ones. Describe, with the exact ping options you would use, how you would confirm or rule out an MTU problem.

The IOS Command Line

EXAM TOPICS **2.4**

🎯 Learning Outcomes

By the end of this chapter you will be able to:

- **Move** between IOS command modes without hesitating, and recognise which mode a prompt indicates.
- **Distinguish** the running configuration from the startup configuration, and save deliberately rather than by habit.
- **Perform** a first-time device setup: hostname, addressing, passwords, SSH and a banner.
- **Filter** show output, which is what makes long output usable under exam time pressure.

🔗 Before You Start

Chapter 4 for addressing and Chapter 6 for the diagnostic commands you will now be typing for real.

📖 Key Terms in This Chapter

user EXEC • privileged EXEC • global configuration • interface configuration • line configuration • running-config • startup-config • NVRAM • context-sensitive help • pipe • banner • SSH

7.1 Modes, and the prompt that tells you where you are

Mode	Prompt	What you can do, and how you got there
User EXEC	SW1>	Look, barely. Enter with enable
Privileged EXEC	SW1#	All show commands, copy, reload, debug. Enter configuration with configure terminal
Global configuration	SW1(config)#	Device-wide settings: hostname, users, routing
Interface configuration	SW1(config-if)#	One interface. Entered with interface Gi0/1
Line configuration	SW1(config-line)#	Console and VTY access. Entered with line vty 0 15
VLAN configuration	SW1(config-vlan)#	Naming a VLAN. Chapter 8
Router configuration	R1(config-router)#	A routing protocol. Chapter 17

💡 Getting out, and the two ways to do it

`exit` steps back one level. `end`, or `Ctrl-Z`, jumps straight to privileged EXEC from anywhere. When an exam question shows you a prompt, the prompt is telling you which commands are currently legal — a global command typed in interface mode is rejected, and that is a favourite trap.

One shortcut worth knowing: `do` runs a privileged EXEC command without leaving configuration mode, so `do show ip interface brief` works from `SW1(config-if)#`.

7.2 Two configurations, and only one survives a reboot

📖 running-config and startup-config

The **running configuration** is in RAM. It is what the device is doing right now, and every command you type takes effect there *immediately*.

The **startup configuration** is in NVRAM. It is what the device will load when it next boots. Nothing you type reaches it until you copy it there.

- `copy running-config startup-config` — save. Abbreviated `wr` or `write` memory on most platforms.
- `show running-config` — what is in force.
- `show startup-config` — what will be in force after a reload.

⚠️ Not saving is a technique, not only a mistake

Because configuration takes effect immediately and is lost on reload, a deliberately unsaved change is a safety net. If you are about to modify the access list or the interface you are connected through, and you get it wrong, a power cycle restores the last saved configuration and your access with it.

The professional habit is: make the risky change, verify you still have access, *then* save. Reversing that order is how people lock themselves out of devices in another building.

7.3 Getting help, and typing less

Keystroke	Effect
<code>?</code>	List every command valid here
<code>sh ?</code>	List commands starting with “sh”
<code>show ip ?</code>	List the next keyword. Note the space — <code>show ip?</code> means something different
Tab	Complete the current keyword
Abbreviation	Type the shortest unambiguous form. <code>sh ip int br</code> is <code>show ip interface brief</code>
<code>Ctrl-A</code> / <code>Ctrl-E</code>	Jump to start / end of line
<code>Ctrl-C</code>	Abandon this command
<code>Ctrl-Shift-6</code>	Abort a running ping or traceroute
Up arrow	Previous command

💡 Filtering output is a survival skill

A full `show running-config` on a production switch runs to hundreds of lines. The pipe cuts it to what you need, and in an exam it is the difference between answering and scrolling.

- `| include vlan` — only lines containing “vlan”
- `| exclude ^!` — drop the `!` comment lines
- `| begin interface` — start output at the first match
- `| section interface Gi0/1` — that interface’s whole block

🔍 SW1# `show running-config | section interface GigabitEthernet0/1`

```
interface GigabitEthernet0/1
description Uplink to SW2
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 10,20,30
```

7.4 A first-time setup, end to end

➤ Configure — a switch from the box to remotely manageable

```
enable
configure terminal
!
! 1. Identify the device. The hostname appears in every prompt and log.
hostname SW1
!
! 2. Protect privileged mode. "secret" is hashed; "password" is not --
!     never use "enable password".
enable secret Str0ng-Enable-Secret
!
! 3. A local user for SSH, with full privilege.
username netadmin privilege 15 secret Str0ng-User-Secret
!
! 4. A management address. A layer 2 switch has no routed ports, so
!     this lives on an SVI, and it needs a gateway to be reachable
!     from another subnet.
interface Vlan1
ip address 10.0.0.11 255.255.255.0
no shutdown
exit
ip default-gateway 10.0.0.1
!
! 5. SSH needs a domain name and a key. 2048-bit minimum; version 2 only.
ip domain-name example.local
crypto key generate rsa modulus 2048
ip ssh version 2
!
! 6. Console: require login, and stop output interrupting your typing.
line console 0
login local
logging synchronous
exec-timeout 10 0
exit
!
! 7. Remote access: SSH only, authenticated against local users.
line vty 0 15
login local
transport input ssh
exec-timeout 10 0
```

```

exit
!
! 8. A legal banner. Never write "welcome" -- it has been argued in court.
banner motd # Authorised access only. Activity is logged. #
!
end
copy running-config startup-config

```

Q SW1# show ip ssh

```

SSH Enabled - version 2.0
Authentication timeout: 120 secs; Authentication retries: 3

```

! Common Pitfalls

- **Using enable password.** It is stored in clear text. `enable secret` is hashed, and where both exist the secret wins.
- **Forgetting transport input ssh.** Leave it at the default and Telnet stays available, which undoes the point of configuring SSH.
- **Generating a key without a hostname and domain name.** The key is named after them, and the command fails or produces a useless key if they are still at defaults.
- **Saving before verifying.** See the warning above. Verify access, then save.
- **Omitting logging synchronous.** Without it a log message arrives in the middle of the line you are typing, and you cannot tell what you have actually entered.

🔧 Break & Fix — The change worked yesterday and the device came back without it.

A colleague configured a new VLAN and confirmed it working. After a power cut overnight, the VLAN is gone and the configuration looks like it did last week.

Diagnosis. The change was made in the running configuration and never copied to the startup configuration. The reload loaded NVRAM, which did not have it. Nothing is broken — the device did exactly what it was told.

Confirm it by comparing the two before doing anything else:

Q SW1# show archive config differences nvram:startup-config system:running-config

```

!Contextual Config Diffs:
+vlan 30
+ name Guest
+interface GigabitEthernet0/9
+ switchport access vlan 30

```

Any output at all means the two differ and a reload will lose the difference.

Fix. Re-apply the change, verify, then `copy running-config startup-config`. As a habit, run that comparison before you leave any device.

🔧 Lab 7.1 — Bring up a switch you can reach from your desk (Packet Tracer)

Topology. One switch, one PC on the same VLAN, connected by console cable and by an Ethernet cable.

Task 1. Connect by console. Move through all five modes in turn and write down the prompt at each. Use `end` to jump back from the deepest.

Task 2. Set hostname, `enable secret` and a local user.

Task 3. Give `Vlan1` an address in the PC's subnet and a default gateway. Ping the switch from the PC to prove it.

Task 4. Enable SSH properly: domain name, 2048-bit key, version 2, and `transport input ssh` on the VTY lines.

Task 5. SSH from the PC. Then try Telnet and confirm it is refused.

Task 6. Make a small change, do *not* save, reload, and observe the change vanish. Repeat with a save.

Task 7. Extension. Practise `| include`, `| section` and `| begin` on `show running-config` until you can extract a single interface block in one command without thinking.

? Checkpoint — Test Yourself

1. You are at `SW1(config-if)#` and need to check the routing table without losing your place. What do you type?
2. Which configuration does a device load on boot, and which one are you editing when you type a command?
3. Why must a hostname and domain name be set before generating an RSA key?

✓ Chapter Summary

- The prompt tells you the mode, and the mode decides which commands are legal. `end` returns to privileged EXEC from anywhere; `do` runs one EXEC command without leaving.
- Commands take effect in the running configuration immediately and are lost on reload until copied to the startup configuration.
- Verify access first, save second. An unsaved risky change is a safety net.
- `enable secret`, not `enable password`. SSH version 2 with a 2048-bit key, and `transport input ssh` to shut off Telnet.
- Filtering with `| include`, `| section` and `| begin` is what makes device output usable at speed.

✍ Review Questions

1. (MCQ) Which prompt indicates global configuration mode? (a) `SW1>` (b) `SW1#` (c) `SW1(config)#` (d) `SW1(config-if)#`
2. (MCQ) Which command shows only the lines of the running configuration that mention a trunk? (a) `show run | begin trunk` (b) `show run | include trunk` (c) `show run | section trunk` (d) `show run trunk`
3. (Short) Explain the difference between `| include` and `| section`, with an example where the answers differ.
4. (Short) Give the professional argument for *not* saving a configuration immediately after making a change to a remote device.
5. (Applied) Write the complete configuration to take a factory-default switch to the point where you can administer it by SSH from another subnet, and state which single omission would leave Telnet available.

PART II

Switching and Network Access

This part and the next are half the exam between them. Domain 2, Switching and Network Access, is 25% on its own, and it is the domain where the blueprint stops asking you to explain things and starts asking you to *configure* them: three of its five topics begin with that word.

It is also the part where a HEC-trained student is furthest from the exam. No published HEC networking course outline in any edition names a VLAN, a trunk, an EtherChannel or a spanning tree. Everything in these seven chapters is new ground, and all of it is typed rather than read.

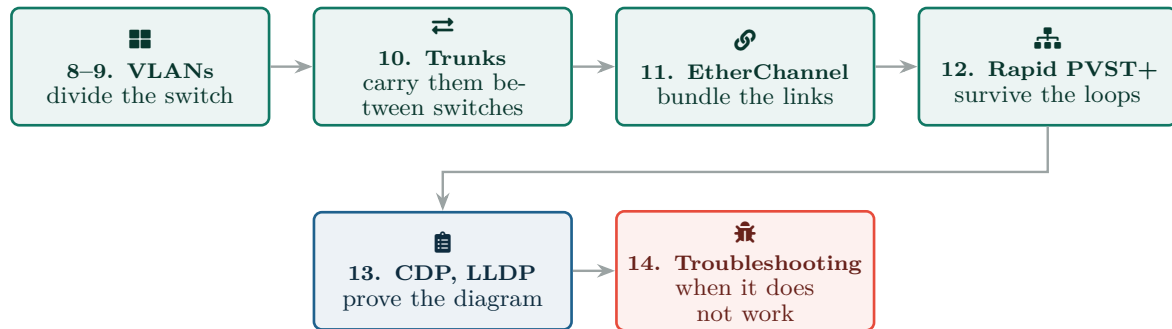


Figure P2.1: Part II. Each chapter adds one capability to the same switched network, and the last two are about proving it works and finding out why it does not.

Where the marks are in domain 2

Topic 2.1 (switch-to-switch and switch-to-router connectivity) and 2.2 (edge-host port attributes) are both *Configure* topics and between them cover trunks, EtherChannel, SVIs, VLANs and PoE. Topic 2.5 is *Configure* as well — Rapid PVST+ with PortFast and the guards. Topic 2.3 asks you to *validate* documentation with CDP and LLDP, and 2.4 to *troubleshoot* with `show` commands, ping, traceroute and capture output.

Four of the five are hands-on. Read this part with a simulator open.

VLANs and Access Ports

EXAM TOPICS **2.2**

🎯 Learning Outcomes

By the end of this chapter you will be able to:

- **Explain** what a VLAN changes about a switch, in terms of the learning-and-flooding algorithm from Chapter 3.
- **Create** VLANs and assign access ports to them.
- **Verify** VLAN membership from three different angles, and know which one to trust when they disagree.
- **Diagnose** the classic VLAN faults: a port in the wrong VLAN, a VLAN that does not exist, and a VLAN that is shut down.

🔗 Before You Start

Chapter 3 for the MAC table and flooding; Chapter 7 for moving around the command line.

📖 Key Terms in This Chapter

VLAN • broadcast domain • access port • VLAN database • default VLAN • native VLAN • management VLAN • SVI • VLAN 1 • shutdown VLAN

8.1 What a VLAN actually does

📖 VLAN

A **VLAN** is a broadcast domain implemented in software on a switch. Ports in VLAN 10 behave as though they were on a physically separate switch from ports in VLAN 20: a broadcast from one never reaches the other, and a frame cannot cross between them without passing through something that routes.

💡 One line changes in the switching algorithm

Recall the algorithm from Chapter 3: learn the source, then forward, flood, or filter. VLANs change exactly one thing — **every step is now scoped to a VLAN**.

- Learning is per VLAN. The same MAC address may legitimately appear in two VLANs on the same switch.
- Flooding is per VLAN. An unknown unicast in VLAN 10 goes out every *VLAN 10* port and no others.
- A frame never leaves the VLAN it arrived in. Not once, not for broadcast, not for anything.

That is the whole mechanism. Everything else about VLANs — trunking, inter-VLAN routing, voice VLANs — is machinery for coping with the consequences of that one rule.

⚠ The consequence people forget

If two devices are in different VLANs they cannot communicate, *even if they are in the same IP subnet, plugged into adjacent ports on the same switch*. The switch will not carry the frame. Conversely two devices in the same VLAN but different subnets cannot communicate either, because IP will not deliver it.

VLAN and subnet are separate ideas that must be kept aligned by design. The convention that saves everyone trouble is one VLAN, one subnet, and a number that matches: VLAN 10 is 192.168.10.0/24, VLAN 20 is 192.168.20.0/24.

8.2 Creating VLANs and assigning ports

➤ Configure — three VLANs and the access ports that use them

```
! Create the VLANs and name them. A name is documentation that
! survives on the device -- use it.
vlan 10
  name Staff
vlan 20
  name Guest
vlan 30
  name Voice
exit
!
! Assign a single access port.
interface GigabitEthernet0/5
  description Staff desk 5
  switchport mode access
  switchport access vlan 10
  no shutdown
!
! Assign a block of ports in one go.
interface range GigabitEthernet0/6 - 12
  description Staff desks 6-12
  switchport mode access
  switchport access vlan 10
!
interface range GigabitEthernet0/13 - 18
  description Guest desks
  switchport mode access
  switchport access vlan 20
```

💡 Why switchport mode access matters

Without it the port is left in dynamic auto mode, which means it will negotiate and may become a *trunk* if the device on the other end asks for one. On an access port that is both a fault waiting to happen and a security hole — an attacker who can negotiate a trunk can reach every VLAN.

Set it explicitly. It is one line, and it is the difference between a port that does what you configured and a port that does what the far end suggests.

8.3 Verifying, three ways

Q SW1# show vlan brief

VLAN	Name	Status	Ports
1	default	active	Gi0/1, Gi0/2, Gi0/3, Gi0/4
10	Staff	active	Gi0/5, Gi0/6, Gi0/7, Gi0/8 Gi0/9, Gi0/10, Gi0/11, Gi0/12
20	Guest	active	Gi0/13, Gi0/14, Gi0/15
30	Voice	active	
1002	fddi-default	act/unsup	

This is the view by VLAN. Note that VLAN 30 exists but has no ports — it was created and never used, which is a common and harmless state, and also a common symptom of a half-finished change.

Q SW1# show interfaces GigabitEthernet0/5 switchport

```
Name: Gi0/5
Switchport: Enabled
Administrative Mode: static access
Operational Mode: static access
Administrative Trunking Encapsulation: dot1q
Access Mode VLAN: 10 (Staff)
Trunking Native Mode VLAN: 1 (default)
Voice VLAN: none
```

This is the view by port, and it is the one to trust when something is odd. **Administrative** mode is what you configured; **operational** mode is what the port has actually become. When those two differ, you have found your fault.

Q SW1# show mac address-table vlan 10

Mac Address Table			
Vlan	Mac Address	Type	Ports
10	000a.f36e.7712	DYNAMIC	Gi0/5
10	0060.5c2b.19a4	DYNAMIC	Gi0/7

This is the view by evidence: which devices are actually present in the VLAN. If a port is in VLAN 10 and no MAC address ever appears against it, nothing is plugged in or the device is silent — and that distinction is the start of most access-layer troubleshooting.

! Common Pitfalls

- **Assigning a port to a VLAN that does not exist.** On many platforms the switch creates it silently; on others the port simply stops forwarding. Either way `show vlan brief` tells you within seconds.
- **Leaving everything in VLAN 1.** VLAN 1 cannot be deleted and carries control traffic. Do not put users on it, and do not use it as the management VLAN.
- **Forgetting that a VLAN can be shut down.** `shutdown` inside `vlan 10` stops the whole VLAN forwarding on that switch while every port still shows as correctly assigned. It is an unusual fault and therefore a nasty one.
- **Deleting a VLAN and leaving ports assigned to it.** Those ports become inactive. `show vlan brief` will not list them at all, which is confusing until you know to look for their absence.

🔧 Break & Fix — A new desk cannot reach anything, and the port looks fine.

A user on Gi0/9 has no connectivity. The port is up, the cable is good, and the configuration reads exactly like the working port next to it.

```
SW1# show vlan brief | include Staff|Gi0/9
```

```
10    Staff                                active    Gi0/5, Gi0/6, Gi0/7, Gi0/8
```

Gi0/9 is not listed against VLAN 10 — nor against any other VLAN.

```
SW1# show interfaces GigabitEthernet0/9 switchport
```

```
Name: Gi0/9
Switchport: Enabled
Administrative Mode: static access
Operational Mode: static access
Access Mode VLAN: 15 (Inactive)
```

Diagnosis. The port was assigned to VLAN 15, which does not exist on this switch — most likely a typing slip for 10, or a VLAN that was deleted after the port was configured. The word (Inactive) is the whole diagnosis: the port is administratively correct and operationally orphaned.

Fix. Either create VLAN 15 if it was meant to exist, or reassign the port: `switchport access vlan 10`. Confirm with `show vlan brief` that Gi0/9 now appears in the Staff list, then confirm a MAC address learns against it.

Lab 8.1 — Divide one switch into three networks (*Packet Tracer*)

Topology. One switch, six PCs — two per VLAN — all in the same IP subnet to begin with.

Task 1. With every port in VLAN 1, confirm all six PCs can ping each other. Record the MAC address table.

Task 2. Create VLANs 10, 20 and 30 with names, and move two PCs into each.

Task 3. Re-test. Which pings now succeed and which fail? Explain each result in terms of broadcast domains, not IP.

Task 4. Run all three verification commands from this chapter and reconcile them.

Task 5. Assign Gi0/11 to VLAN 99, which you have not created. Record what `show vlan brief` and `show interfaces switchport` each say.

Task 6. Enter `vlan 20` and type `shutdown`. Confirm the ports still look correctly assigned while nothing passes. Undo it.

Task 7. Extension. Re-address the PCs so each VLAN is its own subnet. Note that inter-VLAN pings still fail — and that fixing it needs Chapter 10.

Checkpoint — Test Yourself

- Two PCs are in the same IP subnet, plugged into the same switch, in VLANs 10 and 20. Can they ping each other? Justify your answer at layer 2.
- Which command distinguishes what you configured on a port from what the port has actually become?
- A port shows `Access Mode VLAN: 15 (Inactive)`. State the two possible causes.

Chapter Summary

- A VLAN is a broadcast domain in software. Learning, flooding and forwarding all become per-VLAN operations.
- A frame never leaves its VLAN without being routed.

- Keep VLAN and subnet aligned one-to-one; mismatches produce faults that look like everything and nothing.
- Always set `switchport mode access` explicitly, or the port may negotiate a trunk.
- Verify three ways: `show vlan brief` by VLAN, `show interfaces switchport` by port, and the MAC table by evidence. Administrative versus operational mode is where faults show.

Review Questions

1. **(MCQ)** Which command assigns an access port to VLAN 10? (a) `vlan 10` (b) `switchport access vlan 10` (c) `switchport trunk vlan 10` (d) `ip vlan 10`
2. **(MCQ)** A port reports Administrative Mode: `dynamic auto` and Operational Mode: `trunk`. What has happened? (a) it was configured as a trunk (b) it negotiated a trunk with the far end (c) the VLAN is inactive (d) the port is errdisabled
3. **(Short)** Explain why leaving a port in dynamic auto is a security problem as well as an operational one.
4. **(Short)** A VLAN exists, ports are assigned, and nothing forwards. Name a cause that `show vlan brief` would reveal and one it would not.
5. **(Applied)** You are asked to segment a 24-port switch for staff, guests and printers, with guests unable to reach the other two. Write the configuration, and state what additional device or feature you would need for the isolation requirement to hold once routing is added.

Edge-Host Connectivity: PoE, Voice, APs and IoT

EXAM TOPICS **2.2**

🎯 Learning Outcomes

By the end of this chapter you will be able to:

- **Configure** an access port correctly for each of the five device types the blueprint names.
- **Explain** how a data VLAN and a voice VLAN share one cable, and why that is not a trunk in the ordinary sense.
- **Configure** and verify Power over Ethernet, including budget and per-port limits.
- **Diagnose** the faults peculiar to edge ports: a phone that gets no power, an AP that sees only one VLAN, and a device in the wrong VLAN because it was plugged into the wrong socket.

🔗 Before You Start

Chapter 8 for VLANs and access ports. Chapter 7 for interface ranges.

📖 Key Terms in This Chapter

Power over Ethernet • PoE, PoE+, UPoE • power budget • voice VLAN • data VLAN • standalone AP • controller-based AP • LWAPP/CAPWAP • IoT appliance • errdisable

📖 Read topic 2.2 carefully — it lists the devices

Topic 2.2 is “configure Layer 2 switch port attributes for edge-host connectivity (VLAN, PoE, port channel, and LACP)” and then names five categories: desktops, printers and IoT appliances; wireless access points, standalone and controller-based; VoIP phones; virtualised hosts; and network appliances. The exam expects a different port configuration for several of them, so learn them as five cases rather than one.

9.1 Power over Ethernet

📖 PoE

Power over Ethernet delivers electrical power over the same twisted pair that carries data, so a phone, camera or access point needs one cable rather than two. The switch negotiates with the device and supplies only what it asks for.

Standard	Per port	Name	Typically powers
802.3af	15.4 W	PoE	IP phone, basic AP, door sensor
802.3at	30 W	PoE+	Modern AP, PTZ camera
802.3bt	60–90 W	UPoE, PoE++	Multi-radio AP, display, thin client

⚠ The budget is the constraint, not the port

A switch has a total power budget shared across all ports — often far less than the number of ports times the maximum per port. A 24-port PoE+ switch with a 370 W budget cannot deliver 30 W to all 24. When the budget is exhausted the switch stops powering *new* devices, so the symptom is “the phone I plugged in this morning works, the one I plugged in this afternoon does not”, and nothing in the port configuration looks wrong.

Check `show power inline` before you check anything else.

>_ Configure — PoE on an access port

```
interface GigabitEthernet0/10
  description AP in the east corridor
  switchport mode access
  switchport access vlan 10
  ! auto is the default: negotiate and give the device what it asks for.
  power inline auto
  ! Cap a port that must never take more than a phone's worth.
  ! power inline auto max 15400
  no shutdown
  !
  ! Never power this port -- a data-only port that must not energise.
interface GigabitEthernet0/24
  power inline never
```

🔍 SW1# show power inline

Available:370.0(w) Used:71.5(w) Remaining:298.5(w)

Interface	Admin	Oper	Power (Watts)	Device	Class	Max
Gi0/1	auto	on	6.3	IP Phone 7960	2	15.4
Gi0/2	auto	on	6.3	IP Phone 7960	2	15.4
Gi0/10	auto	on	28.6	AIR-AP2802I	4	30.0
Gi0/11	auto	off	0.0	n/a	n/a	30.0
Gi0/24	never	off	0.0	n/a	n/a	0.0

Read the top line first. **Remaining** is what decides whether the next device you plug in will power up at all.

9.2 Voice VLANs: two VLANs, one cable

An IP phone has a small switch inside it: the cable from the wall goes into the phone, and the PC plugs into the back of the phone. Both need network access, and they should not be in the same VLAN — voice needs its own subnet for quality and for security.

💡 How one access port carries two VLANs

The port is configured with a **data VLAN** and a **voice VLAN**. The switch tells the phone, over CDP or LLDP-MED, which VLAN ID to tag its own traffic with. The phone tags voice frames and passes the PC's frames through untagged.

So the switch sees tagged voice traffic and untagged data traffic on one access port. Strictly this is a trunk carrying two VLANs, but it is configured as an access port and Cisco calls it a **multi-VLAN access port**. Do not configure `switchport mode trunk` here — the voice VLAN command does what is needed, and a real trunk would change how the PC's untagged traffic is handled.

>_ Configure — a desk port for a phone with a PC behind it

```

interface GigabitEthernet0/1
  description Desk 1 - phone with PC daisy-chained
  switchport mode access
  switchport access vlan 10
  switchport voice vlan 30
  ! The PC should forward immediately rather than waiting for
  ! spanning tree -- see chapter on Rapid PVST+.
  spanning-tree portfast
  power inline auto
  no shutdown

```

Q SW1# show interfaces GigabitEthernet0/1 switchport

```

Name: Gi0/1
Switchport: Enabled
Administrative Mode: static access
Operational Mode: static access
Access Mode VLAN: 10 (Staff)
Voice VLAN: 30 (Voice)

```

9.3 Access points, standalone and controller-based

The blueprint distinguishes the two, and the distinction changes the port configuration completely.

	Standalone (autonomous)	Controller-based (lightweight)
Where the config lives	On the AP itself	On the wireless LAN controller
What crosses the wire	Each SSID's traffic in its own VLAN	All client traffic tunnelled to the controller in CAPWAP
Switch port	Trunk , carrying every SSID's VLAN	Access port , in the AP management VLAN
Scales to	A handful of APs	Hundreds

⚠ The most common AP misconfiguration

Configuring a lightweight AP's port as a trunk, or a standalone AP's port as an access port. Both leave the AP reachable — it will get an address and appear healthy — while clients on some or all SSIDs get nothing. If wireless users on one SSID work and another SSID does not, look at the switch port before you touch the wireless configuration.

>_ Configure — both kinds of access point

```

! Lightweight AP: one access port in the AP management VLAN.
! Client traffic is tunnelled to the controller, so no client
! VLANs need to exist on this switch at all.
interface GigabitEthernet0/10
  description Lightweight AP - CAPWAP to WLC
  switchport mode access
  switchport access vlan 50
  power inline auto
  spanning-tree portfast
!
! Standalone AP: a trunk, because the AP bridges each SSID
! into a different VLAN and tags them itself.
interface GigabitEthernet0/11
  description Standalone AP - SSIDs in VLANs 10 and 20
  switchport trunk encapsulation dot1q

```

```
switchport mode trunk
switchport trunk native vlan 50
switchport trunk allowed vlan 10,20,50
power inline auto
```

9.4 Printers, IoT appliances and virtualised hosts

- **Printers and IoT appliances** — ordinary access ports. Put them in their own VLAN: they are rarely patched, often speak plaintext protocols, and are a favourite foothold. Add port security (Chapter 25) because they never move.
- **Virtualised hosts** — a server running virtual machines usually needs a *trunk*, because each VM may sit in a different VLAN and the hypervisor’s virtual switch tags accordingly. This is the one “end device” that is not an access port.
- **Network appliances** — firewalls, load balancers and the like. Read the vendor’s requirement; most want a trunk, some want several access ports.

! Common Pitfalls

- **PortFast on a port that leads to a switch.** PortFast skips the spanning tree listening and learning states. On an edge port that is correct; on a port leading to another switch it can create a loop. Use BPDU guard alongside it (Chapter 12).
- **Assuming the phone’s PC port is in the voice VLAN.** It is not — untagged traffic from the PC lands in the data VLAN, which is the whole point.
- **Ignoring the power class.** A class 4 device on a switch with only 802.3af ports will not power up, however correct the configuration looks.
- **Trunking to a lightweight AP.** See the warning above.

🔧 Break & Fix — Half the wireless users work; the other half get no address.

Users on the Staff SSID are fine. Users on Guest associate to the wireless network but never receive an IP address. The AP is reachable and its configuration has not changed.

SW1# show interfaces GigabitEthernet0/11 trunk

Port	Mode	Encapsulation	Status	Native vlan
Gi0/11	on	802.1q	trunking	50

Port	Vlans allowed on trunk
Gi0/11	10,50

Diagnosis. This is a standalone AP, correctly trunked, but the allowed VLAN list carries 10 and 50 only. VLAN 20 — the Guest SSID’s VLAN — is not permitted on the trunk, so Guest clients associate at the radio (which the AP handles alone) and then have no path to the DHCP server. Association succeeding while addressing fails is the signature of a VLAN that does not reach the wire.

Fix. Add the missing VLAN to the trunk: `switchport trunk allowed vlan add 20`. Note add — without it you replace the whole list and drop VLANs 10 and 50, taking down the users who were working. Confirm with `show interfaces trunk`, then have a Guest client renew.

🔧 Lab 9.1 — Wire an edge properly (Packet Tracer)

Topology. One PoE switch; an IP phone with a PC behind it; a lightweight AP; a printer.

Task 1. Create VLANs 10 (Staff), 20 (Guest), 30 (Voice), 50 (AP-Mgmt) and 60 (Printers).

- Task 2.** Configure the phone port with data VLAN 10, voice VLAN 30, PortFast and PoE. Verify both VLANs appear on the one port.
- Task 3.** Configure the AP port as an access port in VLAN 50 with PoE.
- Task 4.** Configure the printer port in VLAN 60 with PortFast.
- Task 5.** Run `show power inline` and record the budget, the used figure and the class of each powered device.
- Task 6.** Set `power inline never` on the printer port and observe what happens to a PoE device moved there.
- Task 7. Extension.** Change the AP to standalone: reconfigure its port as a trunk carrying VLANs 10, 20 and 50, and explain in one sentence why the two AP types need opposite port configurations.

? Checkpoint — Test Yourself

1. An IP phone powers up and works; a PC plugged into the back of it gets an address in VLAN 10. Which VLAN carries the phone's own traffic, and how did the phone learn to use it?
2. A new AP will not power on. `show power inline` shows the port `auto/off` and `Remaining:4.2(w)`. What is wrong?
3. Why does a lightweight AP need only one VLAN on its switch port when it serves four SSIDs?

✓ Chapter Summary

- PoE classes: 15.4 W af, 30 W at, up to 90 W bt. The shared *budget* is what actually runs out, so read `show power inline` first.
- A voice VLAN plus a data VLAN on one access port lets a phone and a PC share a cable: the phone tags, the PC does not.
- Lightweight APs take an access port in the management VLAN; standalone APs take a trunk carrying every SSID's VLAN. Getting this backwards breaks some SSIDs and not others.
- Printers and IoT devices belong in their own VLAN with port security. Virtualised hosts usually need a trunk.
- Use `switchport trunk allowed vlan add`, never bare `allowed vlan`, on a live trunk.

🔧 Review Questions

1. (MCQ) Which port configuration suits a controller-based access point? (a) trunk with all client VLANs (b) access port in the AP management VLAN (c) routed port (d) access port in VLAN 1
2. (MCQ) Which command stops a port ever supplying power? (a) `power inline auto` (b) `power inline never` (c) `no power inline` (d) `shutdown`
3. (Short) Explain why a port with a voice VLAN is not configured as a trunk, even though it carries two VLANs.
4. (Short) A device on a PoE port shows `Oper: off` while other ports are powering normally. Give two causes and the command that distinguishes them.
5. (Applied) You must connect twelve IP phones with PCs behind them, four PoE+ access points and six printers to one 24-port switch with a 370 W budget. Estimate whether the budget suffices, state your assumptions, and write the configuration for one port of each type.

Trunking, 802.1Q and SVIs

EXAM TOPICS **2.1**

🎯 Learning Outcomes

By the end of this chapter you will be able to:

- **Explain** what an 802.1Q tag is and why the native VLAN is the exception that causes most trunk faults.
- **Configure** a trunk between two switches and between a switch and a router, and prune it to the VLANs it should carry.
- **Configure** switch virtual interfaces so that a multilayer switch routes between VLANs.
- **Diagnose** the four trunk faults: mode mismatch, encapsulation mismatch, native VLAN mismatch and a missing VLAN in the allowed list.

🔗 Before You Start

Chapter 8 for VLANs and access ports; Chapter 4 for addressing, which the SVI section needs.

📖 Key Terms in This Chapter

trunk • 802.1Q • tag • native VLAN • allowed VLAN list • DTP • dynamic auto • dynamic desirable • SVI • inter-VLAN routing • router on a stick • subinterface

10.1 The tag, and the one VLAN that has none

A single link between two switches must carry traffic for every VLAN, and the receiving switch has to know which VLAN each frame belongs to. **802.1Q** solves this by inserting a four-byte tag into the frame carrying the VLAN ID.

💡 The native VLAN is untagged, and that is the whole problem

One VLAN on every trunk is designated the **native VLAN**, and its frames cross the trunk **untagged**. The receiving switch assumes any untagged frame it sees on a trunk belongs to *its own* native VLAN.

If the two ends disagree about which VLAN is native, traffic silently leaks between VLANs: frames sent untagged in VLAN 1 by one switch are received into VLAN 99 by the other. Nothing goes down, nothing logs an error at layer 2, and two VLANs that should be isolated are now joined. CDP will complain, which is one of several reasons to leave it on (Chapter 13). Set the native VLAN explicitly and identically at both ends, and make it an unused VLAN — never VLAN 1, and never a VLAN carrying users.

➤ Configure — a trunk between two switches, configured deliberately

```
! On BOTH ends, identically.
interface GigabitEthernet0/1
  description Uplink to SW2
  ! Some platforms require the encapsulation before the mode; on
  ! switches that only support dot1q the command is rejected as
  ! unavailable, which is not an error.
  switchport trunk encapsulation dot1q
```

```

switchport mode trunk
! An unused VLAN as native, matched at the far end.
switchport trunk native vlan 999
! Carry only what this link needs. Pruning limits broadcast
! flooding and limits the blast radius of a mistake.
switchport trunk allowed vlan 10,20,30
! Do not negotiate -- we have decided what this port is.
switchport nonegotiate
no shutdown

```

Q SW1# show interfaces trunk

Port	Mode	Encapsulation	Status	Native vlan
Gi0/1	on	802.1q	trunking	999
Port Gi0/1	Vlans allowed on trunk 10,20,30			
Port Gi0/1	Vlans allowed and active in management domain 10,20,30			
Port Gi0/1	Vlans in spanning tree forwarding state and not pruned 10,20,30			

Four blocks, and they narrow as they go. A VLAN can be *allowed* but not *active* (it does not exist on this switch), or active but not *forwarding* (spanning tree is blocking it). When a VLAN is missing from the last block but present in the first, you have found the layer at which it is being lost.

⚠ allowed vlan replaces; allowed vlan add appends

Typing `switchport trunk allowed vlan 40` on a live trunk does not add VLAN 40 — it makes 40 the *only* allowed VLAN and instantly drops every other VLAN on that link. On a production uplink this is an outage.

Use `switchport trunk allowed vlan add 40`. The equivalents are `remove`, `except` and `all`.

10.2 DTP, and why to turn it off

Dynamic Trunking Protocol lets two Cisco switches negotiate whether a link becomes a trunk. It is convenient and it is a liability.

Local mode	vs trunk	vs dynamic desirable	vs dynamic auto / access
trunk	trunk	trunk	trunk / mismatch
dynamic desirable	trunk	trunk	trunk / access
dynamic auto	trunk	trunk	access / access
access	mismatch	access	access / access

The two entries worth memorising are the failures. `dynamic auto` on both ends gives you an **access port**, not a trunk — each side waits for the other to ask. And `trunk` facing `access` is a genuine mismatch that passes only the access VLAN.

💡 Configure both ends; negotiate nothing

Set `switchport mode trunk` and `switchport nonegotiate` on uplinks, and `switchport mode access` on edge ports. Then a port is what you said it is, and an attacker who plugs a laptop into a wall socket cannot negotiate himself a trunk into every VLAN.

10.3 Routing between VLANs

A VLAN is a broadcast domain and traffic cannot leave it without being routed. There are two ways to provide that, and the exam names the modern one.

10.3.1 Switch virtual interfaces — the modern way

An **SVI** is a virtual layer 3 interface inside a multilayer switch, representing one VLAN. Give each VLAN an SVI with an address and the switch routes between them at hardware speed.

>_ Configure — a multilayer switch routing three VLANs

```
! Routing must be switched on -- easily forgotten, and without it the
! SVIs come up and nothing is routed between them.
ip routing
!
interface Vlan10
  description Staff gateway
  ip address 192.168.10.1 255.255.255.0
  no shutdown
!
interface Vlan20
  description Guest gateway
  ip address 192.168.20.1 255.255.255.0
  no shutdown
!
interface Vlan30
  description Voice gateway
  ip address 192.168.30.1 255.255.255.0
  no shutdown
```

⚠ What it takes to bring an SVI up

An SVI shows up/up only when *all* of the following hold:

- the VLAN exists in the VLAN database and is not shut down;
- at least one access port in that VLAN is up, **or** the VLAN is allowed and forwarding on a trunk;
- the SVI itself is not shut down.

An SVI that stays down with a perfectly good address almost always means the second condition: no live port in that VLAN. It is not an addressing fault.

🔍 SW1# show ip interface brief | include Vlan

Vlan10	192.168.10.1	YES manual up	up
Vlan20	192.168.20.1	YES manual up	up
Vlan30	192.168.30.1	YES manual up	down

VLAN 30's SVI is up administratively and down at protocol level — no active port in VLAN 30. Plug in a phone, or allow VLAN 30 on a trunk, and it comes up.

10.3.2 Router on a stick — the older way

Where the switch cannot route, one router interface carries every VLAN over a trunk, divided into **subinterfaces**.

>_ Configure — router on a stick

```
interface GigabitEthernet0/0
  description Trunk to SW1
  no ip address
```

```

no shutdown
!
interface GigabitEthernet0/0.10
  description Staff
  encapsulation dot1Q 10
  ip address 192.168.10.1 255.255.255.0
!
interface GigabitEthernet0/0.20
  description Guest
  encapsulation dot1Q 20
  ip address 192.168.20.1 255.255.255.0
!
! The native VLAN's subinterface is marked, not tagged.
interface GigabitEthernet0/0.999
  encapsulation dot1Q 999 native

```

Every VLAN's traffic crosses the one physical link twice — in and out — so this scales badly. It appears in the exam because it still appears in small networks.

! Common Pitfalls

- **Native VLAN mismatch.** The two ends disagree, VLANs leak, and nothing goes down. CDP logs it; look for %CDP-4-NATIVE_VLAN_MISMATCH.
- **Bare allowed vlan on a live trunk.** Replaces the list. Always add.
- **Forgetting ip routing on a multilayer switch.** The SVIs come up and traffic still will not cross between VLANs.
- **Expecting an SVI to be up with no active port in its VLAN.** It will not be, and no amount of re-addressing changes that.
- **Leaving VLAN 1 as native.** It carries control traffic and it is every attacker's first guess.

🔧 Break & Fix — One VLAN crosses the uplink; the other three do not.

After a maintenance window, users in VLAN 10 are fine and users in 20, 30 and 40 cannot reach the servers. Both uplink ports show as trunking.

Q SW2# show interfaces trunk

Port	Mode	Encapsulation	Status	Native vlan
Gi0/1	on	802.1q	trunking	999
Port	Vlans allowed on trunk			
Gi0/1	10			

Diagnosis. The trunk is healthy; the allowed list has been reduced to VLAN 10 alone. Somebody ran `switchport trunk allowed vlan 10` intending to add VLAN 10 to the existing list, and replaced it instead. The link stayed up throughout, which is why nothing alerted.

Fix. Restore the full list explicitly rather than adding one VLAN at a time: `switchport trunk allowed vlan 10,20,30,40`. Verify with `show interfaces trunk` on *both* ends — the allowed list is configured per side, and the effective set is the intersection, so a matching mistake at the far end would leave the fault half-fixed and twice as confusing.

🏠 Lab 10.1 — Trunk two switches and route between VLANs (*Packet Tracer*)

Topology. Two switches trunked together; a multilayer switch or router providing gateways; two PCs per VLAN across both switches.

- Task 1.** Create VLANs 10, 20 and 999 on both switches. Put PCs in 10 and 20 on each switch.
- Task 2.** Before configuring the trunk, try to ping between the two switches' VLAN 10 PCs. Record the failure and explain it.
- Task 3.** Configure the trunk on both ends with native VLAN 999 and an allowed list of 10, 20 and 999. Re-test.
- Task 4.** Change the native VLAN on one end only to 998. Record what still works, what breaks, and what the log says.
- Task 5.** Restore it, then run `switchport trunk allowed vlan 20` on one end and observe the outage you have just caused. Fix it with `add`.
- Task 6.** Enable `ip routing` and configure SVIs for VLANs 10 and 20. Prove inter-VLAN routing works.
- Task 7. Extension.** Shut every access port in VLAN 20 and watch the SVI go down. Explain why in terms of the three conditions above.

Checkpoint — Test Yourself

1. Both ends of a link are set to `dynamic auto`. What does the link become, and why?
2. An SVI has a correct address and shows `up/down`. What is the most likely cause and how would you confirm it?
3. Why is a native VLAN mismatch more dangerous than a trunk that fails to come up at all?

Chapter Summary

- 802.1Q tags every VLAN except the native one, which crosses untagged. Mismatched native VLANs leak traffic between VLANs silently.
- Configure both ends explicitly and add `switchport nonegotiate`. `dynamic auto` on both ends yields an access port, not a trunk.
- `allowed vlan` replaces the list; `allowed vlan add` appends. This distinction causes real outages.
- `show interfaces trunk` narrows from `allowed` to `active` to `forwarding`; the block where a VLAN disappears tells you the layer at fault.
- An SVI needs `ip routing`, an existing unshut VLAN, and at least one live port in that VLAN.

Review Questions

1. (MCQ) Which command adds VLAN 40 to a trunk without disturbing the VLANs already carried? (a) `switchport trunk allowed vlan 40` (b) `switchport trunk allowed vlan add 40` (c) `switchport trunk vlan 40` (d) `switchport access vlan 40`
2. (MCQ) A trunk's two ends have different native VLANs. What happens? (a) the trunk fails to form (b) both ends shut down (c) untagged traffic is bridged between two VLANs (d) all VLANs are pruned
3. (Short) Explain why an SVI can show `up/down` when its address and mask are correct.
4. (Short) Give two reasons to set the native VLAN to an unused VLAN rather than leaving it as VLAN 1.

5. **(Applied)** A colleague reports that after adding a new VLAN, three existing VLANs stopped working across an uplink. State what they almost certainly typed, what they should have typed, and how you would verify the repair on both ends.

EtherChannel and LACP

EXAM TOPICS **2.1**

🎯 Learning Outcomes

By the end of this chapter you will be able to:

- **Explain** why bundling links needs a protocol rather than just more cables, in terms of what spanning tree would otherwise do.
- **Configure** a layer 2 EtherChannel with LACP, and a layer 3 EtherChannel between routed ports.
- **Verify** a bundle and read the flags in `show etherchannel summary`.
- **Diagnose** the parameter mismatches that stop a bundle forming — the single most common EtherChannel fault.

🔗 Before You Start

Chapter 10 for trunks and allowed VLAN lists; Chapter 8 for access ports.

📖 Key Terms in This Chapter

EtherChannel • port channel • LACP • PAgP • active • passive • on • load balancing • bundle • suspended port • layer 3 EtherChannel

11.1 Why bundling needs a protocol

Run two cables between a pair of switches and you do not get twice the bandwidth. You get a loop, and spanning tree (Chapter 12) blocks one of them — so you have paid for two links and can use one.

📖 EtherChannel

An **EtherChannel** bundles two to eight physical links into one logical interface, the **port channel**. Spanning tree sees a single link and does not block any member, so all of the bandwidth is usable, and if one member fails the others carry on without a topology change.

💡 Load balancing is per conversation, not per packet

Frames are distributed across members by hashing fields from the frame — source MAC, destination MAC, source and destination IP, or a combination. Every frame in one conversation hashes the same way, so a conversation always takes one member link.

The consequence catches people out: a single large file transfer between two hosts gets *one member's worth* of bandwidth, not the whole bundle. Four 1 Gbps links do not give one server a 4 Gbps transfer. They give you 4 Gbps of aggregate capacity across many conversations, which is a different thing. When somebody complains that a new EtherChannel “did not make the backup faster”, this is why.

11.2 The negotiation protocols

Protocol	Modes	Forms a bundle with	Notes
LACP 802.3ad	active, passive	active-active, active-passive. Not passive-passive	Open standard. Use this. Passive waits to be asked; two waiters never speak
PAgP	desirable, auto	desirable-desirable, desirable-auto. Not auto-auto	Cisco proprietary. Same trap: two auto ends form nothing
Static	on	on-on only	No negotiation and no safety net. A mismatch here creates a loop rather than failing cleanly

⚠ Never mix on with a negotiating mode

on at one end and active at the other does not form a bundle, and worse, the on end starts forwarding on every member immediately while the far end still treats them as individual links. That is a loop, built by hand. Either both ends negotiate, or both ends are on. LACP active at both ends is the right default.

11.3 Configuring a layer 2 EtherChannel

💡 The rule that governs everything: members must match

Every member port in a bundle, *at both ends*, must agree on:

- speed and duplex;
- switch mode — all access, or all trunk;
- if trunking: the same native VLAN and the same allowed VLAN list;
- if access: the same access VLAN.

One port that disagrees is suspended and stays out of the bundle. Configure the port channel interface and let the settings inherit downward, rather than configuring each member separately — that is how you avoid the mismatch in the first place.

>_ Configure — a two-member LACP trunk bundle, both ends

```
! Put the physical members into the channel group first. Creating
! the group creates the Port-channel interface automatically.
interface range GigabitEthernet0/1 - 2
  description Member of Po1 - uplink to SW2
  shutdown
  channel-group 1 mode active
  no shutdown
exit
!
! Now configure the logical interface. Settings applied here
! propagate to every member, which is what keeps them matching.
interface Port-channel1
  description Uplink bundle to SW2
  switchport trunk encapsulation dot1q
  switchport mode trunk
  switchport trunk native vlan 999
  switchport trunk allowed vlan 10,20,30
!
! Hash on both source and destination IP for a better spread than
! the default source-MAC-only on many platforms.
port-channel load-balance src-dst-ip
```

Q SW1# show etherchannel summary

Flags: D - down P - bundled in port-channel
 I - stand-alone s - suspended
 H - Hot-standby (LACP only)
 R - Layer3 S - Layer2
 U - in use f - failed to allocate aggregator
 M - not in use, minimum links not met
 u - unsuitable for bundling
 w - waiting to be aggregated
 d - default port

Number of channel-groups in use: 1
 Number of aggregators: 1

Group	Port-channel	Protocol	Ports
1	Po1(SU)	LACP	Gi0/1(P) Gi0/2(P)

Read the flags. Po1(SU) means layer **S**witched and **U**sed — healthy. Each member showing (P) is bundled. The ones to worry about are (I) stand-alone, (s) suspended and (D) down.

Flag	What it is telling you
(P)	Bundled and forwarding. What you want
(I)	Stand-alone — the port is up but is not in the bundle. Usually the far end is not running a compatible mode
(s)	Suspended — the port's parameters do not match the others. Compare speed, duplex, mode, native VLAN and allowed list
(D)	Down. A physical problem; go back to Chapter 2
(SU)	On the port channel: layer 2, in use
(RU)	On the port channel: layer 3, in use

11.4 A layer 3 EtherChannel

Between two multilayer switches you may want the bundle routed rather than switched.

>_ Configure — a routed bundle

```
interface Port-channel2
description Routed bundle to Core2
! no switchport makes this a routed interface -- do it BEFORE
! adding an address, or the address command is rejected.
no switchport
ip address 10.255.0.1 255.255.255.252
!
interface range GigabitEthernet0/23 - 24
no switchport
channel-group 2 mode active
no shutdown
```

! Common Pitfalls

- **Passive-passive or auto-auto.** Neither end initiates and no bundle forms. The ports stay up as individual links, which spanning tree then blocks — so you get a working network and half the bandwidth you paid for.
- **Mismatched allowed VLAN lists.** A favourite: the bundle forms, one VLAN is missing from one end, and only that VLAN breaks.
- **Configuring members individually.** Configure `interface Port-channel1`; let it inherit.

- **Adding no switchport after the IP address.** On a layer 3 bundle the order matters.
- **Expecting one flow to use the whole bundle.** It never will.

🔧 Break & Fix — The bundle formed with one member instead of four.

A new four-link bundle shows only Gi0/1 bundled; the other three are suspended.

Q SW1# show etherchannel summary | begin Group

Group	Port-channel	Protocol	Ports
1	Po1(SU)	LACP	Gi0/1(P) Gi0/2(s) Gi0/3(s) Gi0/4(s)

Q SW1# show interfaces GigabitEthernet0/2 switchport | include Access|Trunking|Mode

```
Administrative Mode: static access
Operational Mode: static access
Access Mode VLAN: 1 (default)
Trunking Native Mode VLAN: 1 (default)
```

Diagnosis. Gi0/1 was configured as a trunk before being added to the group, and the others were left as access ports in VLAN 1. Members must agree on switch mode, so the three that disagree with the first are suspended. The (s) flag is the whole diagnosis; the second command identifies which parameter differs.

Fix. Do not fix the members one by one — that invites the same fault again. Apply the trunk configuration to **interface Port-channel1** and let it propagate, then confirm all four show (P). If a member still refuses, compare speed and duplex next.

🔧 Lab 11.1 — Bundle four links and break them (*Packet Tracer*)

Topology. Two switches joined by four Ethernet links; one PC on each.

Task 1. Cable all four links with no EtherChannel configured. Run **show spanning-tree** and record how many are forwarding and how many are blocking. Explain the result.

Task 2. Configure Po1 with LACP active at both ends, trunked, native VLAN 999, allowed 10 and 20. Verify with **show etherchannel summary**.

Task 3. Re-run **show spanning-tree**. How many logical links does it now see?

Task 4. Shut one member. Confirm traffic continues and that spanning tree does not reconverge.

Task 5. Set one member's access VLAN to 20 by configuring the physical port directly. Record the flag it earns and explain it.

Task 6. Change one end to **passive** while the other is also **passive**. Record what the bundle does.

Task 7. Extension. Build a layer 3 EtherChannel between two multilayer switches with a /30 and prove you can ping across it.

🔍 Checkpoint — Test Yourself

1. Two switches are configured **channel-group 1 mode passive** at both ends. What happens, and what does spanning tree do about it?
2. A member port shows (s). Name three parameters you would compare and the command that shows them.
3. Why does adding a second member to a bundle not double the speed of a single file copy?

 Chapter Summary

- An EtherChannel makes several links look like one, so spanning tree blocks none of them.
- LACP **active** at both ends is the sensible default. Passive–passive and auto–auto form nothing; on mixed with a negotiating mode builds a loop.
- Members must match on speed, duplex, switch mode, native VLAN and allowed list. Configure the port channel interface and let it inherit.
- Read `show etherchannel summary` by its flags: (P) bundled, (s) suspended for a mismatch, (I) stand-alone, (D) down.
- Load balancing is per conversation. One flow uses one member.

 Review Questions

1. (MCQ) Which LACP mode combination forms a bundle? (a) passive/passive (b) active/passive (c) auto/auto (d) on/active
2. (MCQ) A member shows the (s) flag. What does it mean? (a) it is bundled (b) it is suspended because its parameters differ (c) it is down (d) it is a layer 3 port
3. (Short) Explain why mixing on with active is more dangerous than a combination that simply fails to bundle.
4. (Short) A backup job runs at 1 Gbps across a four-member 1 Gbps bundle. Explain to the person who ordered the extra links why this is expected behaviour and what would actually help.
5. (Applied) Write the full configuration, both ends, for a three-member LACP trunk carrying VLANs 10, 20 and 30 with native VLAN 999, and list the four `show` outputs you would capture to prove it is healthy.

Rapid PVST+ and the Protection Features

EXAM TOPICS 2.5

🎯 Learning Outcomes

By the end of this chapter you will be able to:

- **Explain** what a layer 2 loop does to a network, and why it is worse than any layer 3 failure.
- **Predict** the root bridge, root ports and designated ports for a given topology, and **influence** the outcome deliberately.
- **Configure** Rapid PVST+, root primary and secondary, PortFast, BPDU guard, root guard and loop guard.
- **Diagnose** an unexpected root bridge, an errdisabled port and a port that will not leave blocking.

🔗 Before You Start

Chapter 3 for flooding — which is what makes a loop catastrophic — and Chapter 10 for trunks between switches.

📖 Key Terms in This Chapter

BPDU • root bridge • bridge ID • bridge priority • root port • designated port • alternate port • path cost • Rapid PVST+ • PortFast • BPDU guard • root guard • loop guard • errdisable

12.1 Why a layer 2 loop is fatal

💡 Three things happen at once, and none of them stops

An IP packet carries a TTL that decrements at each router, so a routing loop eventually discards the packet. **An Ethernet frame has no TTL.** A frame caught in a switching loop circulates until somebody unplugs something.

- **Broadcast storm.** A single broadcast is flooded, arrives back, and is flooded again — multiplying without limit until the links saturate.
- **MAC table instability.** The same source address arrives on two ports repeatedly, so the table is rewritten constantly and unicast forwarding becomes unreliable (see the flap message in Chapter 3).
- **Duplicate frame delivery.** Hosts receive multiple copies, which breaks protocols that assume they will not.

The network does not degrade gracefully; it stops, usually within seconds. This is why spanning tree exists and why it is on by default.

12.2 How the tree is chosen

📖 Bridge ID and root election

Every switch has a **bridge ID**: a 4-bit priority, plus a 12-bit extended system ID carrying the VLAN number, plus its MAC address. The switch with the **lowest bridge ID wins** and becomes the **root bridge**.

Default priority is 32768. Because the extended system ID adds the VLAN number, the effective priority in VLAN 10 is $32768 + 10 = 32778$. Priorities are configured in steps of 4096.

⚠️ Left alone, the oldest switch wins

If every switch keeps the default priority, the election is decided by the lowest MAC address — which usually means the oldest device in the building becomes the root of your network. That is almost never where you want traffic to converge. **Always set the root deliberately.**

💡 Then every port takes a role

Once the root is known:

- Each non-root switch picks its **root port** — the one port with the lowest cumulative path cost to the root. It forwards.
- Each segment picks a **designated port** — the port on that segment closest to the root. It forwards.
- Everything else becomes an **alternate** (or backup) port and **blocks**. That is the loop being broken.

Ties are broken by lowest sender bridge ID, then lowest sender port ID.

Link speed	Cost	Note
10 Mbps	100	Legacy
100 Mbps	19	
1 Gbps	4	
10 Gbps	2	Costs are cumulative along the path to the root

12.3 Configuring Rapid PVST+

Rapid PVST+ runs one rapid spanning tree instance per VLAN. It converges in seconds rather than the 30–50 of the original standard, and it is what the blueprint names.

➤ Configure — Rapid PVST+ with a deliberate root

```
! Run rapid per-VLAN spanning tree.
spanning-tree mode rapid-pvst
!
! On the switch that SHOULD be root for these VLANs:
spanning-tree vlan 10,20,30 root primary
!
! On the switch that should take over if it fails:
! spanning-tree vlan 10,20,30 root secondary
!
! The macro above simply sets a priority. You can be explicit:
! spanning-tree vlan 10 priority 4096
```


Q SW1# show spanning-tree vlan 10

```
VLAN0010
Spanning tree enabled protocol rstp
Root ID    Priority    4106
           Address    00d0.58a1.0200
           This bridge is the root
           Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID  Priority    4106 (priority 4096 sys-id-ext 10)
           Address    00d0.58a1.0200

Interface      Role Sts Cost      Prio.Nbr Type
-----
Gi0/1          Desg FWD 4         128.1    P2p
Gi0/2          Desg FWD 4         128.2    P2p
```

Two things to read first: **This bridge is the root** — is it the switch you intended? — and the priority, 4106, which is $4096 + 10$ and confirms the extended system ID is doing what it should.

Q SW2# show spanning-tree vlan 10

```
VLAN0010
Spanning tree enabled protocol rstp
Root ID    Priority    4106
           Address    00d0.58a1.0200
           Cost        4
           Port        1 (GigabitEthernet0/1)

Interface      Role Sts Cost      Prio.Nbr Type
-----
Gi0/1          Root FWD 4         128.1    P2p
Gi0/2          Altn BLK 4         128.2    P2p
```

Gi0/2 is Altn BLK — alternate, blocking. That is the loop being broken, and it is a healthy state, not a fault.

12.4 The protection features

Topic 2.5 names PortFast, root guard, loop guard and BPDU guard explicitly.

Feature	What it does	Put it where
PortFast	Skips listening and learning; the port forwards immediately	Access ports with an end device. Saves 30 s on boot and stops DHCP timing out
BPDU guard	Errdisables the port if a BPDU arrives	The same access ports. A BPDU there means somebody plugged in a switch
Root guard	Ignores superior BPDUs; puts the port in root-inconsistent state	Ports facing downstream switches you do not want becoming root
Loop guard	Blocks a port that stops receiving BPDUs rather than letting it forward	Root and alternate ports on trunks, where a unidirectional link would otherwise create a loop

⚠ PortFast and BPDU guard belong together

PortFast on its own is a loop waiting to happen: it tells a port to forward immediately, so if somebody plugs a switch into it the loop is live before spanning tree can react. BPDU guard is the safety catch — the instant a BPDU arrives, the port is shut. Configure them as a pair, always.

>_ Configure — protection, per port and globally

```

! Per access port.
interface range GigabitEthernet0/5 - 20
  switchport mode access
  switchport access vlan 10
  spanning-tree portfast
  spanning-tree bpduguard enable
!
! Or globally, which applies to every access port automatically --
! far harder to forget on the next port somebody configures.
spanning-tree portfast default
spanning-tree portfast bpduguard default
!
! On a trunk facing a downstream switch that must never be root:
interface GigabitEthernet0/23
  spanning-tree guard root
!
! Loop guard, globally on all point-to-point links.
spanning-tree loopguard default

```

Q SW1# show interfaces status err-disabled

Port	Name	Status	Reason	Err-disabled Vlans
Gi0/7		err-disabled	bpduguard	

💡 Recovering an errdisabled port

Manually: shutdown then no shutdown on the interface — after removing whatever caused it.
Automatically:

>_ Configure — automatic recovery from BPDU guard

```

errdisable recovery cause bpduguard
errdisable recovery interval 300

```

Automatic recovery is a convenience, not a fix. If a rogue switch is still plugged in the port will simply errdisable again every five minutes.

❗ Common Pitfalls

- **Leaving the root election to chance.** The oldest switch wins, and your traffic detours through the wiring closet nobody has visited in ten years.
- **PortFast without BPDU guard.** See above.
- **PortFast on a trunk to another switch.** It is for edge ports. There is a portfast trunk variant for specific cases, but the default answer is no.
- **Treating a blocking port as broken.** Altn BLK is spanning tree working.
- **Forgetting spanning tree is per VLAN.** A change to VLAN 10 does nothing for VLAN 20. Always name the VLANs.

🔧 Break & Fix — After a new switch was added, traffic takes a strange path.

An access switch was installed in a spare closet. Since then, throughput between two core switches has halved and show spanning-tree vlan 10 on the core reports that it is no longer the root.

 CORE1# show spanning-tree vlan 10 | include Root ID|Priority|Address

Root ID	Priority	32778
	Address	0011.2233.4455
	Cost	19

Diagnosis. The root's priority is 32778 — the default of 32768 + 10 — and the MAC address is not the core's. The new access switch shipped with default priority and a lower MAC address than the core, so it won the election. Every VLAN's traffic now converges on a small access switch connected by a 100 Mbps link, which is where the cost of 19 comes from and why throughput halved.

Fix. Two steps, and do both. Set the intended root explicitly on the core: `spanning-tree vlan 10,20,30 root primary`, and the secondary on the other core switch. Then stop it happening again: apply `spanning-tree guard root` on the core ports facing access switches, so a switch advertising a superior BPDU is ignored rather than obeyed.

Lab 12.1 — Steer the tree, then protect it (*Packet Tracer*)

Topology. Three switches cabled in a triangle — deliberately looped — with PCs on two of them.

Task 1. Cable the triangle with no spanning tree changes. Identify the root from `show spanning-tree` and explain *why* that switch won.

Task 2. Record which port on which switch is blocking.

Task 3. Set SW1 as root primary and SW2 as root secondary for VLAN 1. Re-check and record what moved.

Task 4. Disconnect the link between SW1 and SW3. Time how long traffic takes to recover and note which port changed role.

Task 5. Enable PortFast and BPDU guard on an access port, then plug a switch into it. Record the log message and the port status, then recover the port.

Task 6. Enable `spanning-tree guard root` on SW1's port to SW3, then try to make SW3 root by lowering its priority. Record what happens instead.

Task 7. Extension. Time how long the network takes to recover from a link failure under `spanning-tree mode pvst`, then repeat under `rapid-pvst` and compare the two figures.

Checkpoint — Test Yourself

1. Three switches have default priorities and MAC addresses ending ...0100, ...0050 and ...0200. Which is root, and why?
2. A port shows `Altn BLK`. Is this a fault? Justify your answer.
3. Why must PortFast and BPDU guard be configured together?

Chapter Summary

- An Ethernet frame has no TTL, so a layer 2 loop never self-clears. A broadcast storm takes the network down in seconds.
- Lowest bridge ID becomes root; bridge ID is priority plus VLAN plus MAC. Set it deliberately or the oldest switch wins.
- Root port per switch, designated port per segment, everything else blocks.
- Rapid PVST+ runs one rapid instance per VLAN. Configure it, and remember every command is per VLAN.

- PortFast with BPDU guard on access ports; root guard facing downstream switches; loop guard on trunks. Errdisable is recovered by `shutdown/no shutdown` after removing the cause.

Review Questions

1. **(MCQ)** What is the effective priority of a switch in VLAN 20 with a configured priority of 8192? (a) 8192 (b) 8212 (c) 28692 (d) 32788
2. **(MCQ)** Which feature errdisables a port when a BPDU is received? (a) PortFast (b) BPDU guard (c) root guard (d) loop guard
3. **(Short)** Explain why a layer 2 loop is more damaging than a routing loop.
4. **(Short)** A port is in `root-inconsistent` state. Which feature put it there, and what was it protecting against?
5. **(Applied)** You inherit a three-switch network where the root is a random access switch and no protection features are configured. Write the commands to make the core the root, provide a backup root, and prevent recurrence — and say in what order you would apply them on a live network, with a reason.

CDP, LLDP and Documenting a Network

EXAM TOPICS **2.3**

🎯 Learning Outcomes

By the end of this chapter you will be able to:

- **Explain** what discovery protocols advertise and what they cannot tell you.
- **Configure** CDP and LLDP, including disabling them where they should not run.
- **Validate** a network diagram against the devices themselves, which is precisely what topic **2.3** asks.
- **Use** discovery output to find a mismatch — native VLAN, duplex or an undocumented device.

🔗 Before You Start

Chapter 10 for native VLANs, Chapter 2 for duplex, Chapter 7 for filtering output.

📖 Key Terms in This Chapter

CDP • LLDP • LLDP-MED • neighbour • holdtime • advertisement • capability code • platform • local interface • port ID

🎓 The verb here is *validate*

Topic **2.3** is “validate the accuracy of network documentation using CDP and LLDP”. It is not asking you to describe the protocols. It is asking you to take a diagram that claims SW1 Gi0/1 connects to R1 Gi0/0, and prove from the devices whether that is true. Expect to be shown a diagram and some `show cdp neighbors` output that contradicts it.

13.1 Two protocols, one job

	CDP	LLDP
Standard	Cisco proprietary	IEEE 802.1AB, open
Default state	On (most Cisco platforms)	Off (most Cisco platforms)
Timers	Advertise 60 s, hold 180 s	Advertise 30 s, hold 120 s
Sees	Cisco devices	Anything that speaks LLDP, including other vendors, printers and phones
Voice extension	—	LLDP-MED, used to tell phones their voice VLAN

Run both in a mixed estate: CDP for the Cisco gear, LLDP for everything else.

What a neighbour advertisement carries

Device identifier, the local and remote interface, the platform and capabilities, and depending on the protocol and platform: IP address, IOS version, native VLAN, duplex and power requirements.

Crucially, a neighbour is **directly connected at layer 2**. A device two switches away never appears. That is exactly what makes the output usable as evidence of physical cabling.

13.2 Reading the output

SW1# show cdp neighbors

Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone

Device ID	Local Intrfce	Holdtme	Capability	Platform	Port ID
R1	Gig 0/1	142	R S I	C1111	Gig 0/0/0
SW2	Gig 0/2	167	S I	WS-C2960	Gig 0/1
SW3	Gig 0/3	121	S I	WS-C2960	Gig 0/24
AP-EAST	Gig 0/10	155	H	AIR-AP2802	Gig 0

Read it as four columns that matter: **Device ID** (who), **Local Intrfce** (my port), **Port ID** (their port), **Capability** (what kind of device). Those four are the cabling diagram, taken from the devices rather than from a drawing somebody updated in 2019.

SW1# show cdp neighbors detail

```

-----
Device ID: R1
Entry address(es):
  IP address: 10.0.0.1
Platform: Cisco C1111-4P, Capabilities: Router Switch IGMP
Interface: GigabitEthernet0/1, Port ID (outgoing port): GigabitEthernet0/0/0
Holdtime : 142 sec

Version :
Cisco IOS XE Software, Version 17.09.04a

Duplex: full
Native VLAN: 999

```

detail adds the IP address, the software version, the duplex and the native VLAN — which is how a native VLAN mismatch gets caught before it causes a security incident.

Configure — running both protocols, and switching them off where they do not belong

```

! CDP is on by default; LLDP is not.
lldp run
!
! Turn CDP off on a port facing anything you do not control.
! Advertising your platform and IOS version to a guest network,
! an ISP handoff or an untrusted uplink is free reconnaissance.
interface GigabitEthernet0/24
  description ISP handoff - no discovery
  no cdp enable
  no lldp transmit
  no lldp receive
!
! Globally, if policy demands it:
! no cdp run

```

⚠ Do not disable it everywhere on principle

There is a reflex to switch CDP off across the estate “for security”. Weigh it properly: internally, CDP and LLDP are how you validate cabling, how phones learn their voice VLAN (Chapter 9), and how native VLAN and duplex mismatches are detected. Disable them on untrusted edges — ISP links, guest ports, DMZ segments — and leave them running inside.

13.3 Validating a diagram

📄 Worked Example: The diagram says one thing, the switch says another

The documentation claims:

- SW1 Gi0/2 → SW2 Gi0/1
- SW1 Gi0/3 → SW3 Gi0/1

The output above shows Gi0/2 to SW2 Gi0/1 — correct — but Gi0/3 to SW3 Gig 0/24, not Gi0/1.

What that means in practice. Somebody re-patched SW3 and did not update the diagram. It may be harmless. It may not: if SW3 Gi0/24 is configured as an access port while Gi0/1 was the trunk, the uplink is carrying one VLAN instead of all of them, and the fault will surface later as “the new VLAN does not work on the third floor”.

What to do. Correct the diagram, then check that the port actually in use carries the configuration the design intended. Discovery output tells you what is plugged in; it does not tell you whether the port is configured correctly. That is a second check.

❗ Common Pitfalls

- **Expecting to see non-adjacent devices.** Neighbours are one hop. A device that does not appear may simply be further away.
- **Concluding a device is absent when CDP is off.** A non-Cisco switch, or one with CDP disabled, is invisible to CDP and may still be there. Cross-check with LLDP and the MAC address table.
- **Trusting stale entries.** Holdtime counts down from 180 s. An entry can persist for three minutes after a device is unplugged.
- **Forgetting lldp run.** Without it, `show lldp neighbors` is empty and you may wrongly conclude nothing is connected.
- **Reading the port columns backwards.** Local Intrfce is yours; Port ID is theirs.

🔧 Break & Fix — An undocumented switch appears in the discovery table.

An audit of SW1 shows a neighbour that is not on any diagram:

```
SW1# show cdp neighbors GigabitEthernet0/14 detail
```

```
-----
Device ID: Switch
Entry address(es):
Platform: cisco WS-C2960-8TC-L, Capabilities: Switch IGMP
Interface: GigabitEthernet0/14, Port ID (outgoing port): FastEthernet0/1
Holdtime : 131 sec

Version :
Cisco IOS Software, C2960 Software, Version 12.2(55)SE
```

Diagnosis. An eight-port switch with the default hostname `Switch`, no management address, and a decade-old IOS version, connected to an access port — somebody has plugged a desk

switch into a wall socket to get more ports. It is unmanaged, it participates in spanning tree, and it can influence the root election (Chapter 12).

Fix. Short term: confirm what is downstream of it before unplugging anything, because there are probably users behind it. Long term, this is exactly what `spanning-tree bpduguard` on access ports is for — with it configured, the port would have errdisabled the moment the switch was connected and the problem would have announced itself instead of being found in an audit months later.

Lab 13.1 — Draw the network from the devices (*Packet Tracer*)

Topology. Four switches and a router, cabled by your instructor or a partner without telling you how, plus a deliberately wrong diagram.

Task 1. Using only `show cdp neighbors` from each device, draw the physical topology. Do not look at the cables.

Task 2. Compare your drawing with the supplied diagram and list every discrepancy.

Task 3. Run `show cdp neighbors detail` on one link and record the IOS version, duplex and native VLAN of the neighbour.

Task 4. Create a native VLAN mismatch on one trunk and find the resulting log message.

Task 5. Enable `lldp run` everywhere and compare `show lldp neighbors` with the CDP view. Account for any difference.

Task 6. Disable CDP on one port and confirm the neighbour disappears — noting how long it takes.

Task 7. Extension. Write a short procedure a junior technician could follow to validate a patching change using discovery output alone.

Checkpoint — Test Yourself

1. `show cdp neighbors` lists a device you know is three switches away. What must be true?
2. You unplug a device and it still appears in the table. Why, and for how long?
3. Which single command would tell you a neighbour's IOS version and native VLAN?

Chapter Summary

- CDP is Cisco and on by default; LLDP is open and off by default. Run both in a mixed estate.
- Neighbours are directly connected only — which is what makes the output evidence of cabling.
- Local `Intf` is your port, Port ID is theirs. Those two columns plus the device name are the diagram.
- `detail` adds IP, version, duplex and native VLAN, and catches mismatches before they cause incidents.
- Disable discovery on untrusted edges, not everywhere. Holdtime means entries persist for minutes after a device is removed.

Review Questions

1. (MCQ) Which command enables LLDP globally? (a) `lldp enable` (b) `lldp run` (c) `lldp transmit` (d) `no lldp disable`
2. (MCQ) In `show cdp neighbors`, which column names the port on the *remote* device? (a) Local `Intf` (b) Port ID (c) Platform (d) Capability

3. **(Short)** Give two reasons to leave CDP running inside a network and one reason to disable it at an edge.
4. **(Short)** A diagram says two switches are connected. Neither appears in the other's CDP table. Give three explanations.
5. **(Applied)** You are handed a network with no documentation. Describe a repeatable procedure, naming the commands, to produce an accurate physical topology diagram and state two things your procedure cannot establish.

Troubleshooting Layer 2

EXAM TOPICS **2.4**

🎯 Learning Outcomes

By the end of this chapter you will be able to:

- **Apply** a repeatable order of checks to a layer 2 fault instead of guessing.
- **Select** the right `show` command for a symptom, and read what it returns.
- **Use** `show logging`, ping, extended ping and capture output together to localise a fault.
- **Diagnose** the six faults that account for most switched-network problems.

🔗 Before You Start

All of Part II. This chapter assembles Chapter 8 to Chapter 13 into a method.

📖 Key Terms in This Chapter

symptom • hypothesis • scope • errdisable • show logging • severity • divide and conquer • bottom-up • baseline

🎓 This chapter is the exam's heaviest verb

Topic **2.4** — “troubleshoot basic Layer 2/Layer 3 connectivity and device operations using `show` commands (including `show logs`), ping, extended ping, trace route, and packet capture output” — is one of eight *troubleshoot* topics. Together with the eight *configure* topics they are 55% of the paper. The method below is worth more marks than any single technology in this book.

14.1 Scope first, then layers

💡 Ask “how many?” before you ask “why?”

The single most useful question is how much is broken, because it tells you where to look before you have run a command.

- **One device** → that port, that cable, that host's configuration.
- **One VLAN, everywhere** → the VLAN itself: does it exist, is it shut, is it allowed on the trunks, does its SVI have an address?
- **Everything on one switch** → that switch's uplink, or spanning tree.
- **Everything, everywhere** → a loop, the root bridge, or a routing or gateway failure.
- **Intermittent, under load** → physical layer, duplex, or a saturated link. Almost never a VLAN.

💡 Then work bottom-up

Once you know the scope, walk the layers from the bottom. Each step is one command and rules out a whole class of cause.

1. **Is the port up?** `show interfaces status`
2. **Is it clean?** `show interfaces` — errors, duplex, speed

3. Is it in the right VLAN? `show interfaces switchport`
4. Does the switch see the device? `show mac address-table`
5. Does the VLAN reach the next switch? `show interfaces trunk`
6. Is spanning tree forwarding? `show spanning-tree vlan`
7. Is there a gateway? `show ip interface brief`

Resist starting at step 7 because the user said “I cannot reach the internet”.

14.2 The first three commands

Q SW1# `show interfaces status`

Port	Name	Status	Vlan	Duplex	Speed	Type
Gi0/1	Uplink to SW2	connected	trunk	a-full	a-1000	10/100/1000
Gi0/5	Desk 5	connected	10	a-full	a-100	10/100/1000
Gi0/6	Desk 6	notconnect	10	auto	auto	10/100/1000
Gi0/7	Desk 7	err-disabled	10	auto	auto	10/100/1000
Gi0/8	Desk 8	connected	15	a-full	a-1000	10/100/1000

One screen, four different faults visible. Gi0/6 `notconnect` — nothing plugged in, or a dead cable. Gi0/7 `err-disabled` — a protection feature fired; find out which. Gi0/8 is in VLAN 15, which may or may not be intended — compare against the design.

Q SW1# `show logging | include LINK|LINEPROTO|ERR|SPANTREE`

```
%LINK-3-UPDOWN: Interface GigabitEthernet0/6, changed state to down
%SPANTREE-2-BLOCK_BPDUGUARD: Received BPDU on port Gi0/7 with BPDU Guard
enabled. Disabling port.
%PM-4-ERR_DISABLE: bpduguard error detected on Gi0/7, putting Gi0/7 in
err-disabled state
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on
GigabitEthernet0/1 (999), with SW2 GigabitEthernet0/1 (1).
```

Read the log before you read anything else

The log frequently contains the answer in plain English, timestamped. Above, it names the errdisable cause and a native VLAN mismatch nobody had reported — a fault that is silently bridging VLAN 999 and VLAN 1 between two switches.

Filter it. `show logging | include` with the keywords above turns hundreds of lines into the handful that matter.

14.3 The six faults, and what each looks like

Fault	Symptom	Command that proves it
Port in the wrong VLAN	One user isolated; can reach some hosts, not others	<code>show interfaces switchport</code> — check Access Mode VLAN
VLAN missing from a trunk	One VLAN fails between switches; others fine	<code>show interfaces trunk</code> — compare allowed lists at both ends
Native VLAN mismatch	Nothing obviously down; VLANs leak	<code>show cdp neighbors detail</code> , or the log message above
Errdisabled port	One port dead after somebody plugged something in	<code>show interfaces status</code> <code>err-disabled</code>

Fault	Symptom	Command that proves it
Spanning tree blocking	A path that should work does not; redundant link idle	<code>show spanning-tree vlan N</code> — look for Altn BLK
Duplex or speed mismatch	Slow, not down; errors climbing under load	<code>show interfaces</code> — collisions, late collisions, CRC

Worked Example: Working a fault end to end

Symptom. “Nobody in accounts can reach the file server since this morning. Sales are fine.”

Scope. One VLAN, multiple users, more than one switch. That rules out a single cable and points at the VLAN or its path.

1. `show interfaces status` on the access switch — accounts ports are connected in VLAN 30. Not a port problem.
2. `show mac address-table vlan 30` — addresses are being learned locally. The devices are alive and the switch sees them.
3. `show interfaces trunk` on the uplink — VLAN 30 appears in “allowed” but *not* in “forwarding and not pruned”.
4. `show spanning-tree vlan 30` — the uplink is Altn BLK.
5. `show logging | include SPANTREE` — a topology change overnight; a second switch was added and won the root election for VLAN 30.

Cause. Not a VLAN fault at all. The new switch became root for VLAN 30 only, which moved the tree and put the accounts uplink into blocking. Sales, in VLAN 20, still had the old root and were unaffected — which is exactly why only one VLAN broke.

Fix. Set the intended root explicitly for all VLANs, add root guard on the ports facing access switches, and confirm the uplink returns to Desg FWD.

Common Pitfalls

- **Changing something before you have measured.** If you reconfigure before capturing the current state you can no longer tell whether you fixed it or moved it.
- **Changing more than one thing at once.** Then you do not know which change worked, and you cannot write it up.
- **Believing the reporter’s diagnosis.** “The switch is broken” usually means “my thing does not work”. Take the symptom, discard the theory.
- **Ignoring what still works.** The working half of the network is evidence. Sales working and accounts failing localised the fault above faster than any command did.
- **Skipping the log.** It is timestamped and often explicit.

Break & Fix — Two symptoms, one cause, and a misleading first look.

Users report intermittent slowness. One user cannot connect at all. Both started after a contractor visited the comms room.

 SW1# `show interfaces status | exclude connected`

Port	Name	Status	Vlan	Duplex	Speed	Type
Gi0/7	Desk 7	err-disabled	10	auto	auto	10/100/1000

```
Q SW1# show logging | include ERR_DISABLE|MACFLAP
```

```
%PM-4-ERR_DISABLE: bpduguard error detected on Gi0/7, putting Gi0/7 in
err-disable state
%SW_MATM-4-MACFLAP_NOTIF: Host 0011.2233.4455 in vlan 10 is flapping
between port Gi0/7 and port Gi0/12
```

Diagnosis. Both symptoms are one event. The contractor plugged an unmanaged switch into Gi0/7 and, to “make sure it worked”, also patched it into Gi0/12. That created a loop: the MAC flap is the loop being seen from two ports, and the intermittent slowness is the resulting flooding. BPDU guard then errdisabled Gi0/7, which is the one port that is definitively dead — and which, by shutting, largely saved the network.

Fix. Remove the second patch lead. Confirm the flapping stops. Then recover the port with `shutdown / no shutdown` — *after* the cause is gone, not before, or it will simply errdisable again. Finally, note that BPDU guard did its job: without it this would have been a full broadcast storm rather than one dead port.

Lab 14.1 — Fix five faults against the clock (*Packet Tracer*)

Topology. Three switches, one router, six PCs across three VLANs — built, working, and then broken by a partner.

- Task 1.** Build the network and confirm it works fully. **Capture a baseline:** `show interfaces status`, `show interfaces trunk`, `show spanning-tree vlan` and `show ip interface brief` from every device. You cannot troubleshoot without knowing what right looks like.
- Task 2.** Hand the network to a partner, who introduces **five** faults from the table above without telling you which.
- Task 3.** Find and fix them, writing down for each: the symptom, the scope, the command that proved it, and the fix. Give yourself twenty minutes.
- Task 4.** Swap roles.
- Task 5.** Compare notes: which fault took longest, and would starting with scope rather than commands have found it faster?
- Task 6. Extension.** Introduce a fault that produces *no* log message — a wrong access VLAN is a good one — and discuss why the log is a first stop rather than a complete answer.

Checkpoint — Test Yourself

1. Users on one VLAN across three switches lose connectivity; other VLANs are fine. What are the two most likely causes, and which command separates them?
2. A port is errdisabled. You issue `no shutdown` and it errdisables again within seconds. What have you forgotten?
3. Why does knowing what *still works* narrow a fault faster than knowing what is broken?

Chapter Summary

- Establish scope first — one device, one VLAN, one switch, or everything. Scope points at the layer before you type anything.
- Then work bottom-up: port up, port clean, right VLAN, MAC learned, VLAN on the trunk, spanning tree forwarding, gateway present.
- Read `show logging` early. It is timestamped and often explicit.

- Six faults cover most cases: wrong VLAN, VLAN missing from a trunk, native VLAN mismatch, errdisable, spanning tree blocking, duplex mismatch.
- Capture a baseline while the network works. Change one thing at a time, and remove the cause before recovering an errdisabled port.

Review Questions

1. **(MCQ)** A single VLAN fails between two switches while others work. Which command should you run first? (a) `show ip route` (b) `show interfaces trunk` (c) `show version` (d) `show mac address-table`
2. **(MCQ)** Which status means a protection feature has shut a port? (a) notconnect (b) disabled (c) err-disabled (d) monitoring
3. **(Short)** Explain why establishing scope before running commands makes troubleshooting faster, with an example.
4. **(Short)** Give one fault from this chapter that produces a clear log message and one that produces none, and say how you would find the second.
5. **(Applied)** Write a troubleshooting procedure of no more than eight steps that a first-line technician could follow for “a user cannot reach the network”, naming the command at each step and the decision it resolves.

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